

ये बड़ा ही मजेदार chapter है जो mains or Adv दोनों में खूब पूछा जाता है.... बस difference इतना है अब तक जो हमने physics पढ़ी है उसकी analysis हमने body को particle मानकर किया है actual shape and size को हम COM and rotation में consider करते हैं let's start



Centre of mass is an imaginary point of any extended body or a system of particles which will perform same motion as a particle of same mass would have performed under applications of same external conditions.

वैसे तो COM की actual feel तुम्हें धीरे-धीरे आएगी लेकिन starting के लिए ऐसे समझो अगर तुमने किसी एक cricket ball को spin कराने के लिए घुमा कर फेंका तो ball के surface का हर एक point बड़ा अजीब सा उटपटांग motion करेगा infect अंदर वाले point भी but इन सभी में ball का centre का motion बिलकुल parabolic होगा जैसा हमने kinematics में पढ़ा था just like motion of a particle thrown in projectile इसी special point को COM कहते हैं।



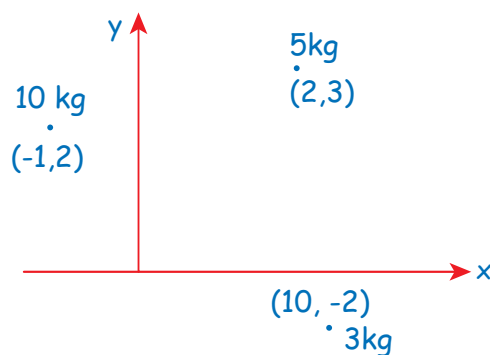
LOCATION OF C.O.M FOR DISCRETE PARTICLE SYSTEM

$$x_{\text{com}} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 \dots}{m_1 + m_2 + m_3 + \dots} = \frac{\sum m_i x_i}{\sum m_i}$$

$$y_{\text{com}} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{m_1 + m_2 + m_3} = \frac{\sum m_i y_i}{\sum m_i}$$

$$z_{\text{com}} = \frac{m_1 z_1 + m_2 z_2 + m_3 z_3}{m_1 + m_2 + m_3} = \frac{\sum m_i z_i}{\sum m_i}$$

Q. Find location of COM of following system.

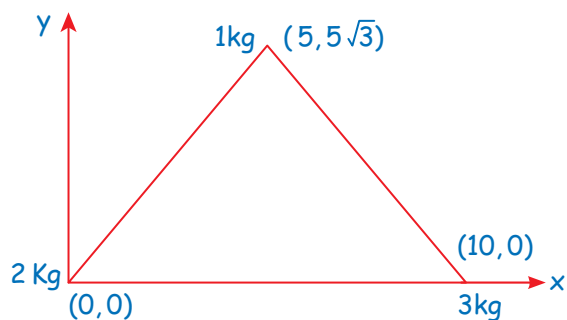


$$\text{Sol. } x_{\text{com}} = \frac{(10 \times (-1)) + 5 \times 2 + 3 \times 10}{10 + 8} = \frac{30}{18}$$

$$y_{\text{com}} = \frac{20 + 15 - 6}{18} = \frac{29}{18}$$

$$\text{com} = \left(\frac{30}{18}, \frac{29}{18} \right)$$

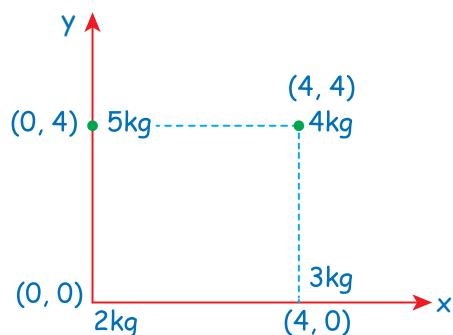
Q. Find location of COM of following system.



Sol. (f) $x_{\text{com}} = \frac{0 + 5 + 30}{6} = \frac{35}{6}$

$y_{\text{com}} = \frac{0 + 5\sqrt{3} + 0}{6} = \frac{5\sqrt{3}}{6}$

Q. Find COM.



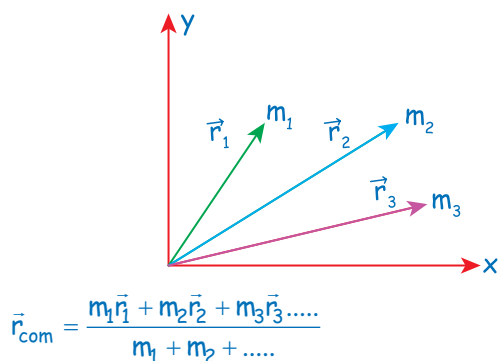
Sol. $x_{\text{com}} = \frac{(2 \times 0) + (3 \times 4) + (4 \times 4) + (5 \times 0)}{5 + 4 + 2 + 3}$

$= \frac{12 + 16}{14} = 2$

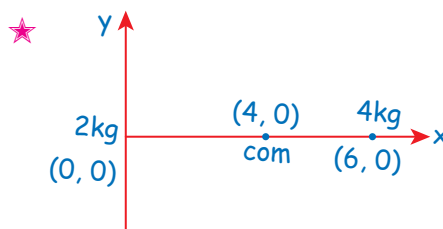
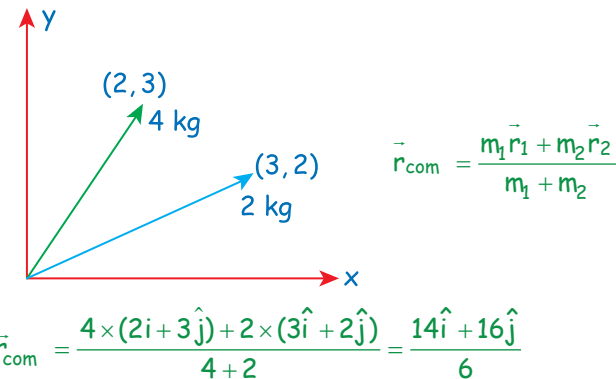
$y_{\text{com}} = \frac{(2 \times 0) + (3 \times 0) + (4 \times 4) + (5 \times 4)}{5 + 4 + 2 + 3}$

$\frac{36}{14} = \frac{18}{7}$

COM OF DISCRETE PARTICLE (IN VECTOR FORM)

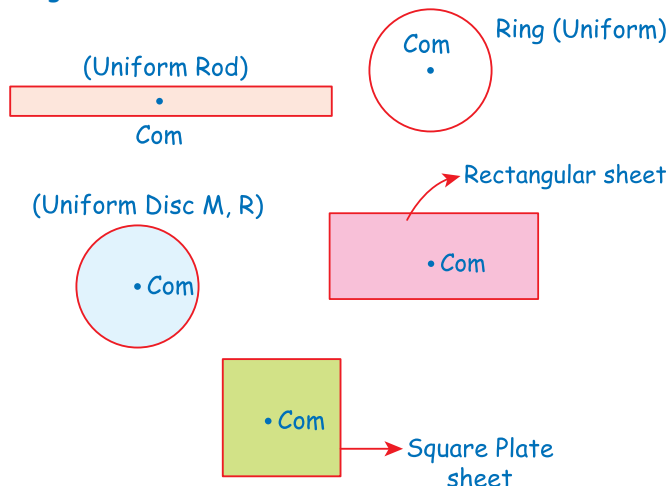


Q. Find location of COM of following system.



★ COM of two particle system will lie on the line joining the two particle system will be close to heavier mass.

★ COM of uniform symmetric body lie on their geometrical centre.

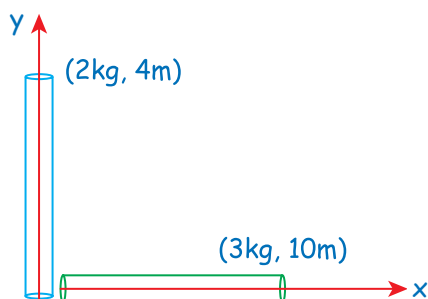


COM OF CONTINUOUS MASS SYSTEM

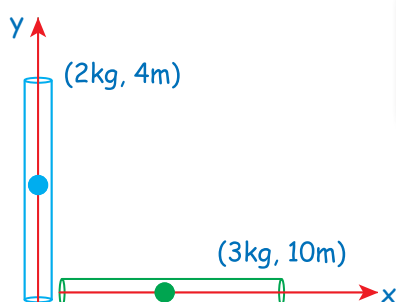
$x_{\text{com}} = \frac{\int x \, dm}{\int dm}$ x-coordinate of COM of dm

$y_{\text{com}} = \frac{\int y \, dm}{\int dm}$ y-coordinate of COM of dm

Q. Find location of COM if both rod are uniform.



Sol.



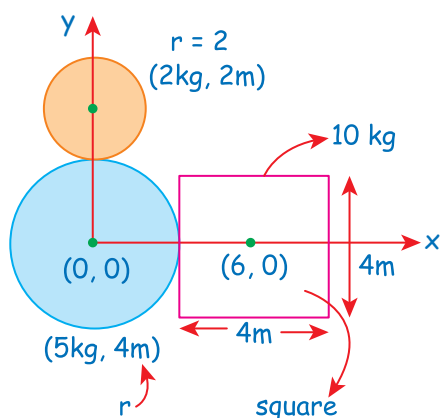
SKC
body hatao
bindu banao



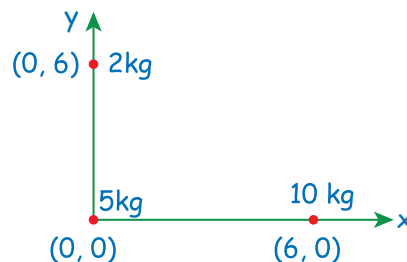
$$x_{\text{com}} = \frac{(3 \times 5) + (2 \times 0)}{2 + 3} = 3$$

$$y_{\text{com}} = \frac{0 + 2 \times 2}{2 + 3} = \left(\frac{4}{5}\right)$$

Q. Find location of COM of following system.



Sol.

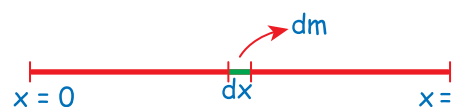


$$x_{\text{com}} = \frac{0 + 0 + 60}{(m_1 + m_2 + m_3)17} = \frac{60}{17}$$

$$y_{\text{com}} = \frac{12 + 0 + 0}{17} = \frac{12}{17}$$

MASS DISTRIBUTION

(1) Linear mass density = λ = mass per unit length



$$dm = \lambda dx$$

ye dx length

ka chota sa mass dm hai

$$\lambda = \frac{dm}{dx}$$

$$M_{\text{total}} = \int dm = \int_{x=0}^{x=l} \lambda dx.$$

If λ is uniform

$$m_{\text{Total}} = \lambda \int_0^l dx$$

$$m = \lambda(l - 0)$$

$$\lambda = \frac{m}{L}$$

(2) Area mass density σ (sheet, Disc)

σ = mass per unit Area

$$\sigma = \frac{dm}{dA}$$

$$dm = \sigma dA$$

$$m_{\text{total}} = \int \sigma dA$$

If σ is uniform

$$M_{\text{Total}} = \sigma A$$

$$\sigma = \frac{m}{A}$$

(3) Volume mass density (ρ)

ρ = mass per volume

$$\rho = \frac{dm}{dv}$$

$$dm = \rho dv$$

$$m_{\text{total}} = m = \int \rho dv$$

If $\rho \rightarrow$ Uniform

$$m_{\text{Total}} = \rho \text{ vol}^n$$

$$\rho = \frac{\text{mass}}{\text{volume}}$$

Q. A rod of length 10 m is placed on x-axis such that it's one end is at origin and another at (10, 0) if linear mass density λ varies as $\lambda = 4x^2$.

Find (a) total mass of rod (b) Location of COM.

Sol. (a) $M_{\text{Total}} = \int \lambda dx = \int_0^{10} 4x^2 dx = \frac{4000}{3}$

(b) $x_{\text{com}} = \frac{\int x dm}{\int dm} = \frac{\int x \lambda dx}{m_{\text{total}}}$

$x_{\text{com}} = \frac{\int_0^{10} x \cdot 4x^2 dx}{\frac{4000}{3}} = \frac{4 \times \frac{10000}{4}}{\frac{4000}{3}} = \frac{15}{2}$

Q. Repeat the above problem if length of rod is l and $\lambda = \lambda_0 x$

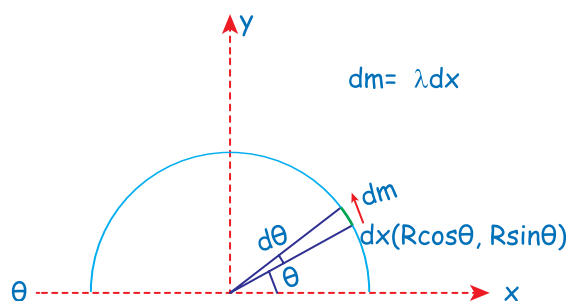
Sol. $M_{\text{Total}} = \int \lambda dx = \int_0^l \lambda_0 x dx = \frac{\lambda_0 l^2}{2}$

$x_{\text{rod}} = \frac{\int x dm}{\int dm}$

$x_{\text{com}} = \frac{\int_0^l x \lambda dx}{\int_0^l \lambda dx} = \frac{\int_0^l x \lambda_0 x dx}{\int_0^l \lambda_0 x dx} = \frac{\frac{l^2}{3}}{\frac{l^2}{2}} = \frac{2l}{3}$

COM OF SEMI RING (M, R)

$x_{\text{com}} = 0$



$\Rightarrow dx = R d\theta \Rightarrow dm = \lambda R d\theta$

$y_{\text{com}} = \frac{\int y dm}{\int dm} = \frac{\int_0^\pi R \sin \theta \cdot \lambda R d\theta}{\int_0^\pi \lambda R d\theta}$

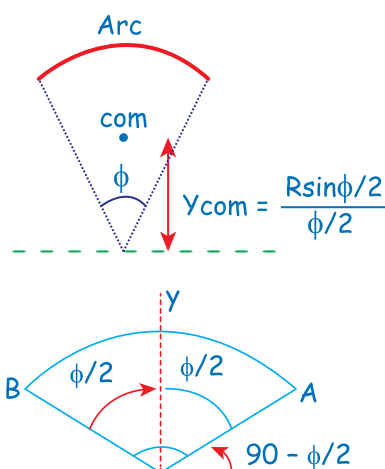
$= \frac{\lambda R^2 \cdot \int_0^\pi \sin \theta d\theta}{\lambda R \int_0^\pi d\theta} = \frac{R(-\cos \theta)|_0^\pi}{(\pi - 0)}$

$\frac{R}{\pi} (\cos \pi - \cos 0) = \frac{-R}{\pi} (-1 - 1) = \frac{2R}{\pi}$

θ पर जाके छोटा सा angle $d\theta$ पकड़ा जिसके सामने की arc length dx है such that $d\theta = dx/R \Rightarrow dx = R d\theta$ (using $\theta = \text{arc/radius}$)



COM OF CIRCULAR ARC OF (RADIUS R)



इसका derivation almost same as for semi ring बस only limit change. अगर हम ध्यान से देखे तो $(\pi/2 - \pi/2)$ से $(\pi/2 + \pi/2)$ तक जाना है



$y_{\text{com}} = \frac{\int y dm}{\int dm} = \frac{\int_A^B R \sin \theta \lambda R d\theta}{\int_A^B \lambda R d\theta}$

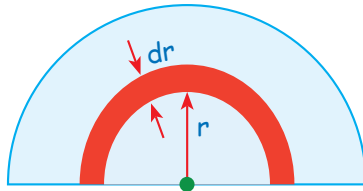
$= \frac{\int_{\pi/2 - \phi/2}^{\pi/2 + \phi/2} R \sin \theta \lambda R d\theta}{\int_{\pi/2 - \phi/2}^{\pi/2 + \phi/2} \lambda R d\theta} = \frac{R \sin(\phi/2)}{\phi/2}$

Q. Find location of com.



Sol. $\phi = 120^\circ \Rightarrow y_{\text{com}} = \frac{R \sin(\pi/3)}{\pi/3} = \frac{3\sqrt{3} R}{2\pi}$

COM OF SEMI DISC



यहाँ हमने r पर जाके dr thickness का एक छोटा सा semi ring पकड़ा जिसका area we can write $dA = \pi r dr$
 $dm = \sigma dA = \sigma \pi r dr$
 This is very very very imp concept

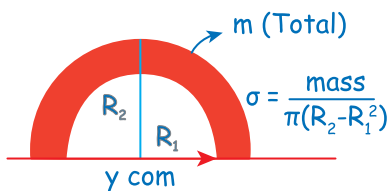


$$x_{com} = 0$$

$$y_{com} = \frac{\int y dm}{\int dm} = \frac{\int_0^R \frac{2r}{\pi} \cdot \sigma \pi r dr}{\int_0^R \sigma \pi r dr}$$

$$= \frac{2 \int_0^R r^2 dr}{\pi \int_0^R r dr} = \frac{2}{\pi} \frac{R^3}{\frac{3R^2}{2}} = \boxed{\frac{4R}{3\pi}}$$

Q. Find com to following strip.



Sol. $y_{com} = \frac{\int_{R_1}^{R_2} \frac{2r}{\pi} \pi r d\sigma}{\int_{R_1}^{R_2} \pi r d\sigma}$

$$= \frac{4(R_1^2 + R_1 R_2 + R_2^2)}{3\pi(R_1 + R_2)}$$

भाई ये भी तो बिलकुल semi disc जैसा ही ques है बस limit R_1 से R_2 करदो... बस इतना ही... है।

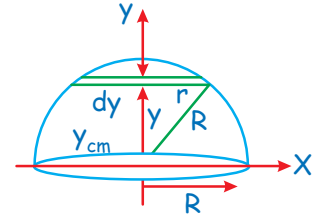
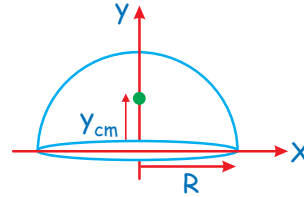


Solid hemisphere, hollow hemisphere and solid cone के COM find out करने का derivation नीचे attach कर रहा हूँ जो कि important नहीं है बस इनके result याद रख लेना।

Center of Mass of a Solid Hemisphere

Let's consider an element disc of width dy , mass dm at a distance y from the center of the hemisphere. The radius of this elemental disc will be given as,

$$r = \sqrt{R^2 - y^2}$$



The mass dm of this disc can be given as

$$dm = \frac{3M}{2\pi R^3} \times \pi r^2 dy = \frac{3M}{2R^3} (R^2 - y^2) dy$$

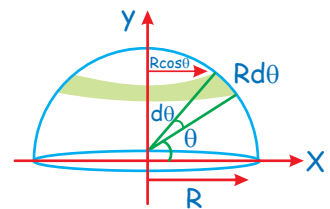
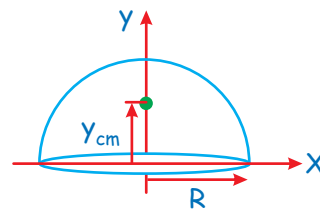
$$y_{cm} = \frac{1}{M} \int_0^R dm y = \frac{1}{M} \int_0^R \left[\frac{3M}{2R^3} (R^2 - y^2) dy \right] y$$

$$\Rightarrow y_{cm} = \frac{3R}{8} \text{ (After solving)}$$

Center of Mass of a Hollow Hemisphere (M, R)

Let's consider an elemental circular strip of angular width $d\theta$ at an angular distance θ from the base of the hemisphere. This strip will have an area.

$$dS = 2\pi R \cos \theta R d\theta$$



$$dm = \frac{M}{2\pi R^2} 2\pi R \cos \theta R d\theta$$

Here y - coordinate of this strip of mass dm can be taken as $R \sin \theta$.

$$y_{cm} = \frac{1}{M} \int_0^{\frac{\pi}{2}} dm R \sin \theta$$

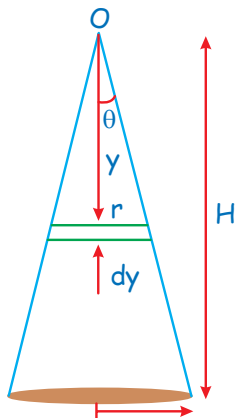
$$= \frac{1}{M} \int_0^{\frac{\pi}{2}} \left(\frac{M}{2\pi R^2} 2\pi R^2 \cos \theta d\theta \right) R \sin \theta$$

$$= R \int_0^{\frac{\pi}{2}} \sin \theta \cos \theta d\theta \Rightarrow y_{cm} = \frac{R}{2}$$

Center of Mass of a Solid Cone

A solid cone has mass M , height H and base radius R . Let's consider an elemental disc of width dy and radius r , at a distance y from the apex of the cone. Let the mass of this disc be dm ,

$$dm = \frac{3M}{\pi R^2 H} \times \pi r^2 dy$$



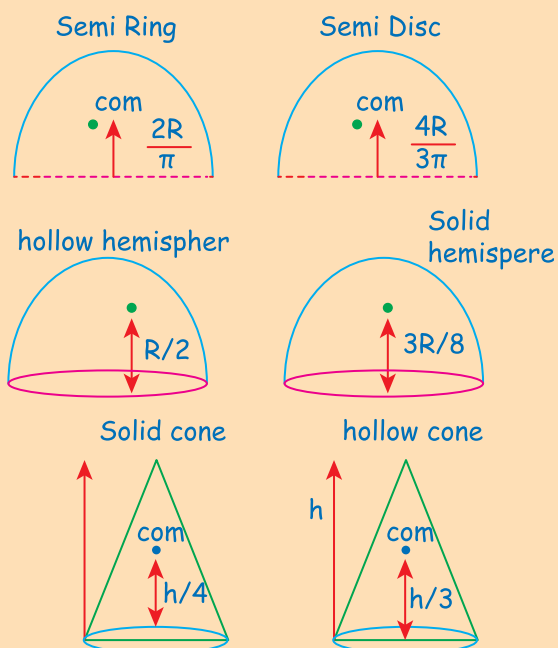
$$y_{cm} = \frac{1}{M} \int_0^H y dm = \frac{1}{M} \int_0^H \left(\frac{3M}{\pi R^2 H} \pi \left(\frac{Ry}{H} \right)^2 dy \right) y$$

$$= \frac{3}{H^3} \int_0^H y^3 dy = \frac{3H}{4} \text{ (from vertex)}$$

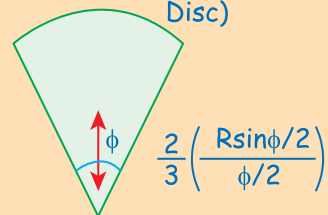


काम का डब्बा

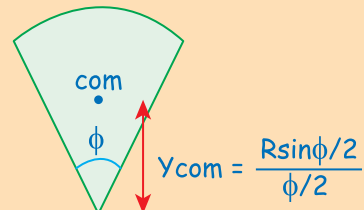
★ Exam से पहले ये जरूर देख कर जाना



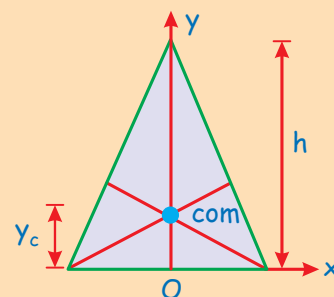
Sector (Part of Disc)



Arc

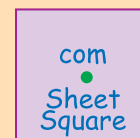


★ A triangular plate (By qualitative argument)

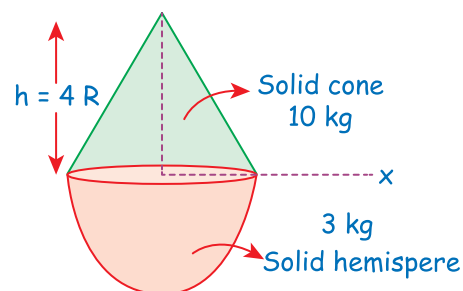


at the centroid: $y_c = \frac{h}{3}$

Ring, Disc, solid and hollow sphere



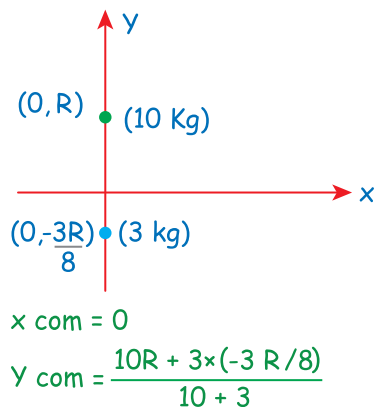
Q. Find location of COM



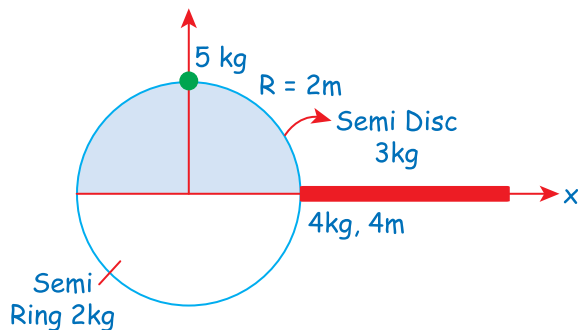
ऐसे ques का एक ही इलाज, body हटाओ बिंदु बनाओ मतलब body के mass को उसके centre of mass पर point mass बनाके solve करलो।



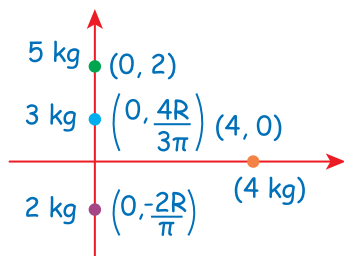
Sol.



Q. In following system we have a point mass of 5 kg, semi disc of 3 kg, semi ring of 2 kg and a uniform rod of length 4 m and mass 4 kg. Find location of COM ($R = 2$ m).



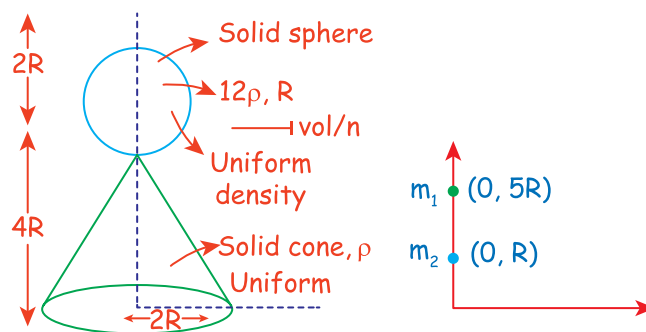
Sol. 4 kg rod has centre at (4, 0)



$$x_{com} = \frac{0 + 0 + 0 + (4 \times 4)}{14}$$

$$y_{com} = \frac{5R + \frac{3 \times 4R}{3\pi} - \frac{2 \times 2R}{\pi} + 0}{14} = \frac{5}{7}$$

Q. Find location of COM (density of sphere is 12ρ & that for cone is ρ)



$$\text{Sol. } m_1 = 12\rho \frac{4}{3}\pi R^3$$

$$m_2 = \rho \cdot \frac{1}{3}\pi(2R)^2 4R = \frac{16\pi R^3 \rho}{3}$$

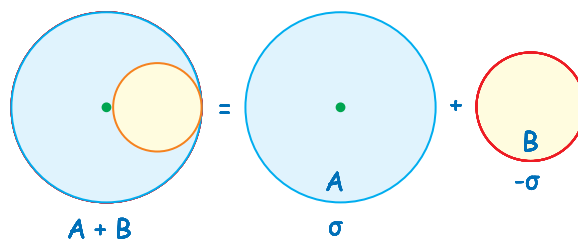
If $m_2 = m$

$$m_1 = 3m$$

$$y_{com} = \frac{m_1 \times 5R + m_2 \cdot R}{m_1 + m_2}$$

$$= \frac{3m \times 5R + mR}{m + 3m} = \frac{16R}{4} = 4R$$

Q. From a disc (m, R) another disc of Radius is removed as shown in diagram. Find location of centre of mass after removing small Disc.



मान लो एक -ve mass ki disc B, badi vali disc A ke upar raki hai to ye ques wali body 2 disc se milkar bani hai

$$m_A = \sigma \pi R^2 = m (\text{let})$$

$$m_B = -\sigma \pi \left(\frac{R}{2}\right)^2 = \frac{-m}{4}$$

$$\text{Sol. } x_{com} = \frac{m \times 0 + \frac{-m}{4} \times \frac{R}{2}}{m - \frac{m}{4}} = \frac{-R}{6}$$

$$com \Rightarrow \left(\frac{-R}{6}, 0\right)$$

MOTION OF COM

यहाँ तक हमने location of COM निकालना सीख लिया अब COM के motion को पढ़ेंगे नीचे मैं COM के लिए बहुत important result लिख रहा हूँ इन्हें अच्छे से practice कर लो क्योंकि पुरा COM इन्हीं पर टिका है।



काम का डब्बा

★ Location of COM = $\vec{r}_{com} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots}{m_1 + m_2 + \dots}$

★ Displacement of COM = $\vec{d}_{com} = \frac{m_1 \vec{d}_1 + m_2 \vec{d}_2 + \dots}{m_1 + m_2 + \dots}$

★ $\vec{v}_{com} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots}{m_1 + m_2} = \frac{\vec{P}_{net}}{m_{total}}$

★ $\vec{P}_{net} = \vec{P} = m_{total} \vec{v}_{com}$ (VVV Important)

★ $\vec{a}_{com} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2 + \dots}{m_1 + m_2} = \frac{(\vec{F}_{ext})_{net}}{m_{total}}$

★ $(\vec{F}_{net})_{ext} = M_{total} \vec{a}_{com}$ (VVV Important)

★ $\frac{d\vec{p}_{net}}{dt} = (\vec{F}_{net})_{ext} = M_{total} \vec{a}_{com}$

★ If $(\vec{F}_{net})_{ext} = 0 \Rightarrow \vec{a}_{com} = 0$

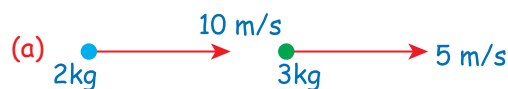
$\vec{v}_{com} = \text{const}$ and momentum of system will conserve

★ If centre of mass is initially at rest and $\sum \vec{F}_{external} = 0$ on centre of mass, then displacement of centre of Mass will be zero. Hence shift in position of centre of mass is zero.

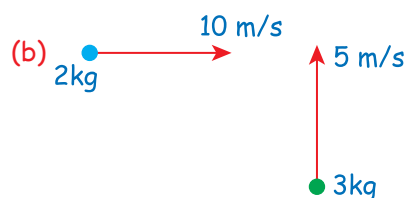
★ If $\sum \vec{F}_{ext}$ is zero on COM then $\vec{v}_{com} = \text{constant}$ but kinetic energy of system may or may not be constant because kinetic energy of the system can also change due to internal forces.

ये बहुत जरूरी है की आप velocity of COM and acc. of COM निकालना सीखें so let's practice it.

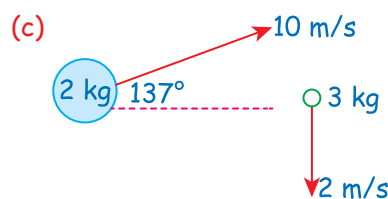
Q. Find velocity of COM in following case.



$$V_{com} = \frac{(2 \times 10) + (3 \times 5)}{2 + 3} = 7 \text{ m/s}$$

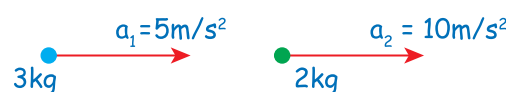


$$V_{com} = \frac{(2 \times 10)\hat{i} + (3 \times 5)\hat{j}}{2 + 3} = 4\hat{i} + 3\hat{j}$$



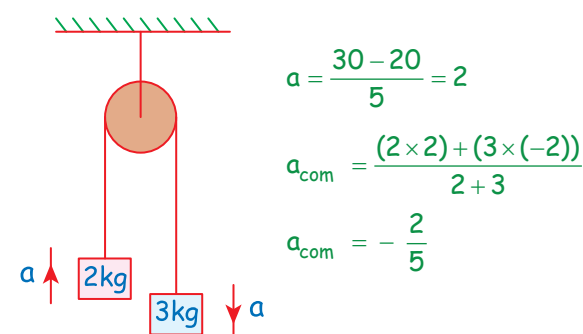
$$V_{com} = \frac{2(8\hat{i} + 6\hat{j}) + 3(-2\hat{j})}{5} = \frac{16\hat{i} - 6\hat{j}}{5}$$

(d) Find acc. of COM.



$$\vec{a}_{com} = \frac{(5 \times 3) + (10 \times 2)}{5} = 7 \text{ m/s}^2$$

(e) Find acceleration of com.



If $(\vec{F}_{\text{net}})_{\text{ext}} = 0$

$\vec{P} \rightarrow$ Conserve

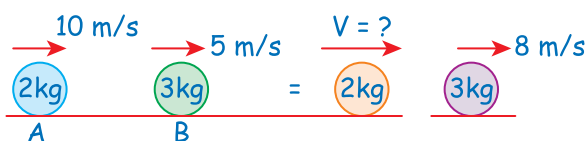
$\vec{P}_i = \vec{P}_f$

This is very important concept बस ये ध्यान रखना आप किसे system मान रहे हो। जिस system पर net ext. force 0 होगा उसका momentum conserve बोले तो const. होगा।

Let's practice ques. on this from basic to adv.

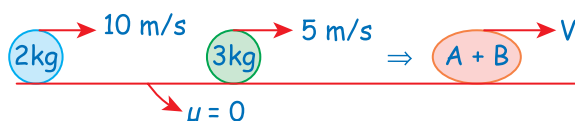


Q. Find velocity(v) to 2kg block after collision.



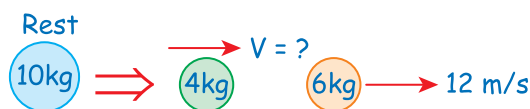
Sol. $P_i = P_f$ $(F_{\text{net ext}})_{A+B} = 0$
 $2 \times 10 + 3 \times 5 = 2V + 3 \times 8$
 $V = 5.5$

Q. 2 kg and 3 kg ball strike and stick together. Find velocity of them after chipka-chipki.



Sol. $P_i = P_f$
 $2 \times 10 + 3 \times 5 = 5 \times V$
 $V = 7 \text{ m/s}$

Q. 10 kg block explode into two fragments 4 kg and 6 kg after explosion 6 kg move with velocity 12 m/s along x-axis, answer following parts.



Sol. (1) Velocity of 4 kg block after explosion?
 $\rightarrow P_i = P_f$
 $0 = 4v + 6 \times 12$
 $\vec{V} = -18 \rightarrow \text{Backward}$
 (2) V_{com} before and after explosion
 $V_{\text{com}} = 0 = \text{constant}$
 Bcz No external force

(3) Total energy released in explosion $KE_f - KE_i$

$$KE_i = 0$$

$$KE_f = \frac{1}{2} \times 4 \times 18^2 + \frac{1}{2} \times 6 \times (12)^2$$

$$648 + 432 = 1080$$

Q. 10kg mass explode as shown in figure. Find velocity of (2kg) block.



Sol. Velocity of (2Kg)block

$$P_i = P_f$$

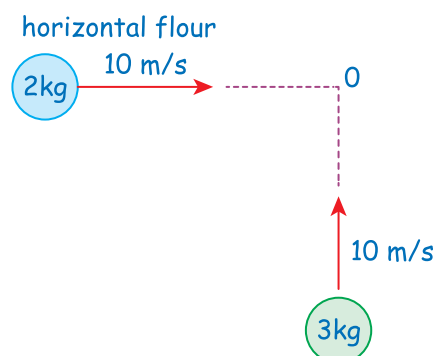
$$0 = 2v + (3 \times 10)\hat{j} + (8 \times 5)\hat{i}$$

$$V = -20\hat{i} - 15\hat{j}$$

Exam में कभी-कभी silly mistake करने लिए speed पूछ लेता है we carefull यहाँ speed of 2 kg is 25 m/s after explosion.



Q. On frictionless horizontal floor two balls are moving as shown in figure.



If both collide at 'O' and stick together Find velocity of (A + B) Just after sticking. and Energy loss in collision.

Sol. Let's after chipka-chipki (sticking) both are moving with velocity V

$$(2 \times 10)\hat{i} + (3 \times 10)\hat{j} = 5\vec{V}$$

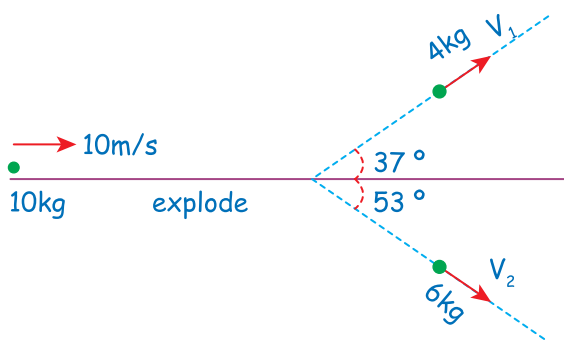
$$\Rightarrow \vec{V} = 4\hat{i} + 6\hat{j}$$

$$(K \cdot E)_i = \frac{1}{2} \times 2 \times 10^2 + \frac{1}{2} \times 3 \times 10^2 = 250$$

$$(K \cdot E)_f = \frac{1}{2} \times 5 \times (4^2 + 6^2) = 130$$

$$\Delta KE = K \cdot E_f - K \cdot E_i = 130 - 250 = -120 \quad (\text{Loss})$$

Q. On horizontal floor 10 kg shell explode in 2 parts as shown. Find speed of 4 kg and 6 kg.



Sol. Momentum conservation in x

$$P_i = P_f : x \text{ में}$$

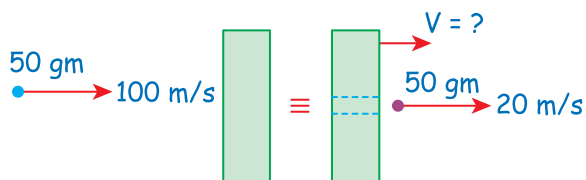
$$10 \times 10 = (4 \times V_1 \cos 37^\circ) + 6(V_2 \cos 53^\circ)$$

momentum conservation in y

$$P_i = P_f : y \text{ में}$$

$$0 = (4V_1 \sin 37^\circ) - (6V_2 \sin 53^\circ) \text{ now you can solve and get the ans.}$$

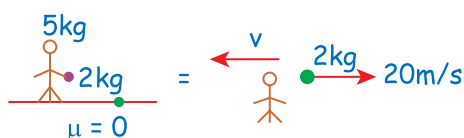
Q. A 50 gm bullet strikes a block and pan through it with velocity 20m/s. Find V.



Sol. Initial momentum $P_i = P_f$ final momentum

$$\frac{50}{1000} \times 100 + 0 = 50v + \frac{50}{1000} \times 20 \Rightarrow V = 0.8 \text{ m/s}$$

Q. Man throws 2kg body as shown in figure. Find velocity of man.



$$\text{Sol. } (\vec{F}_{\text{net}})_{\text{ext}} = 0$$

$$P_i = P_f$$

$$0 = (2 \times 20) - 50 \times V$$

$$V = \frac{40}{50} = \frac{4}{5}$$

Pistol / बंदूक / muzzle velocity / recoil velocity वाले ques में same concept होता है बस आदमी के हाथ में बंदूक होती है। Same concept will use in nuclear physics in 12th class.

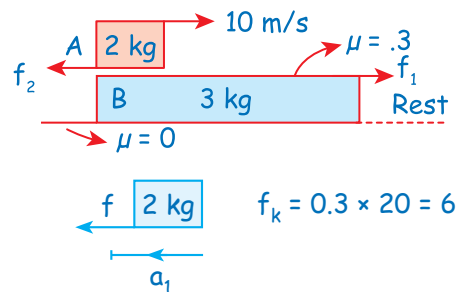


Q. Block A is given velocity 10 m/s as shown above. Answer following questions.

(1) Find when Relative motion stop, or sleeping stop.

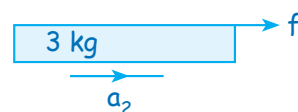
(2) (V_A and V_B) sleeping stop or find common V.

Sol.



$$a_1 = \frac{f}{m} = \frac{6}{2}$$

$$a_1 = 3 \text{ (पीछे)}$$



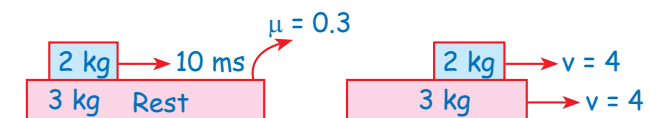
$$a_2 = f_k = \frac{6}{3} = 2$$

Let both have same velocity at time t

$$V = 10 - 3t$$

$$V = 0 + 2t$$

$$t = 2 \text{ sec, } v = 4 \text{ m/s}$$



$$P_i = P_f$$

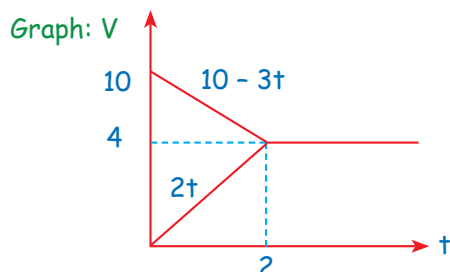
[Momentum conservation]

$$2 \times 10 + 0 = 2V + 3V$$

$$V = 4 \text{ m/s}$$

$$V_{\text{com}} = 0$$

$$V_{\text{com}} = \frac{2 \times 10 + 3 \times 0}{2 + 3} = 4 \text{ m/s हमें था}$$



$$(KE)_i = \frac{1}{2} \times 2 \times 10^2 + 0 = 100 \text{ J}$$

$$(KE)_f = \frac{1}{2} \times 2 \times 4^2 + \frac{1}{2} \times 3 \times 4^2 = 40 \text{ J}$$

Find the work done by fk:

$$W_g + W_N + W_f = \Delta KE$$

$$0 + 0 + W_f = 40 - 100 = -60$$

SKC

★ WD by kinetic friction can be +ve, -ve or zero but net WD by kinetic friction is always -ve.

★ WD by static friction can be +ve, -ve or zero but net WD by static friction is always zero.

★ WD by internal force are frame independent



Impulse (J)

$$J = \int F \cdot dt$$

★ Since, $F = \frac{dp}{dt}$

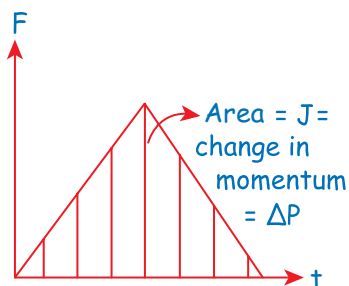
$$\int_{P_i}^{P_f} dp = \int_{t=1}^{t=2} F \cdot dt$$

$$\Delta P = P_f - P_i = \Delta \vec{P} = \int_{t_1}^{t_2} \vec{F} \cdot dt$$

★ $\Delta P = J = \text{Area under } F - t \text{ graph}$

★ $\Delta \vec{P} = \vec{J} = \text{Impulse}$

★ Impulse = change in momentum



Q. Find the impulse due to the force $\vec{F} = a\hat{i} + bt\hat{j}$, where $a = 2 \text{ N}$ and $b = 4 \text{ N/s}$, if this force acts from $t_i = 0$ to $t_f = 0.3 \text{ s}$.

Sol. $\vec{I} = \int_{t_i}^{t_f} \vec{F} dt = \int_0^{0.3} (a\hat{i} + bt\hat{j}) dt$

or $\vec{I} = a\hat{i} \int_0^{0.3} dt + b\hat{j} \int_0^{0.3} t dt$

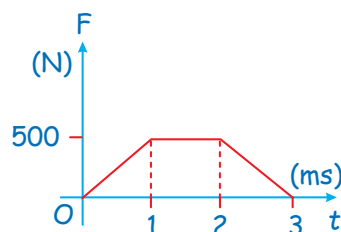
or $\vec{I} = at\hat{i} \Big|_0^{0.3} + \frac{bt^2}{2} \hat{j} \Big|_0^{0.3}$

or $\vec{I} = (2)(0.3)\hat{i} + \frac{(4)(0.3)^2}{2} \hat{j}$

thus $\vec{I} = (0.6\hat{i} + 0.18\hat{j}) \text{ Ns}$

Q. Figure shows the variation of force acting on a body with time. Calculate the impulse of this force.

Sol. Impulse of a force is the area under the graph.



Thus,

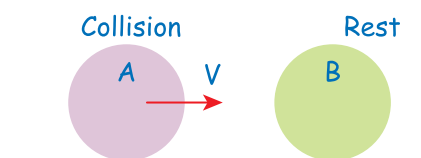
$$I = \left[\frac{1}{2} (500)(1) + (500)(2-1) + \frac{1}{2} (500)(3-2) \right] \times 10^{-3}$$

or $I = [250 + 500 + 250] \times 10^{-3}$

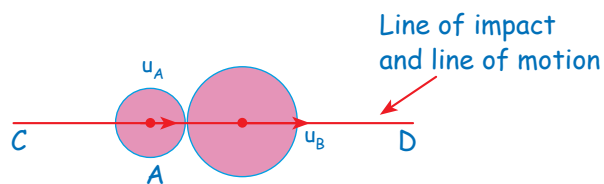
or $I = 1 \text{ Ns}$

COLLISION

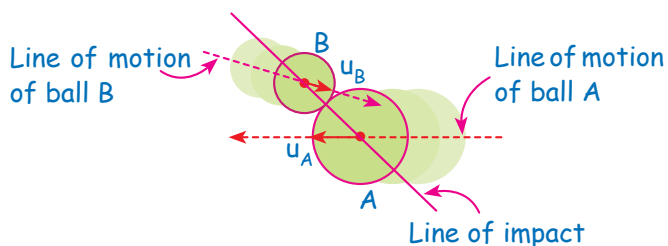
The line passing through the common normal to the surfaces in contact during impact is called line of impact or line of collision (LOI/LOC).



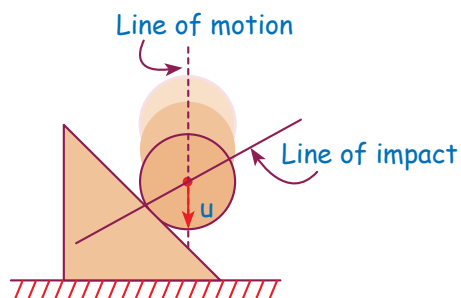
अगर body की velocities before and after the collision LOC की तरफ है तो it's called head on collision, else oblique collision.



Head on Collision



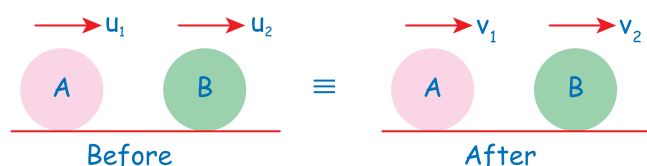
Oblique Collision



Oblique Collision

$$e = \frac{\text{velocity of separation of points of contact along line of impact}}{\text{velocity of approach of points of contact along line of impact}}$$

Head on Collision



$$P_{\text{before}} = P_{\text{after}}$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\text{Coeff of restitution } e = \frac{v_2 - v_1}{u_1 - u_2}$$

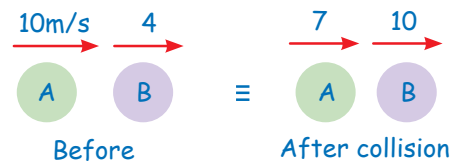
if ($e = 1$) (elastic collision / Perfectly elastic Collision)

if ($0 < e < 1$) (Inelastic collision)

if ($e = 0$) (Perfectly inelastic collision) (stick)

चलो सबसे पहले हम e की definition लगाना सीखते हैं।

Q. Find $e = ?$



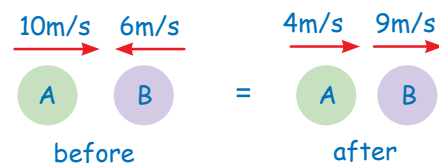
Find $e = ?$

$$\text{Sol. } e = \frac{\text{velocity of sep}}{\text{velocity of app}} = \frac{3}{6} = \frac{1}{2}$$

$$\text{OR } e = \frac{(V_{\text{relative}})_f}{(V_{\text{relative}})_i} = \frac{3}{6} = \frac{1}{2}$$

$$\text{OR } e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{10 - 7}{10 - 4} = \frac{1}{2}$$

Q. Find $e = ?$



$$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{9 - 4}{10 - (-6)} = \frac{5}{16}$$

$$\text{Sol. } e = \frac{5}{16}$$

Q. Find $e = ?$



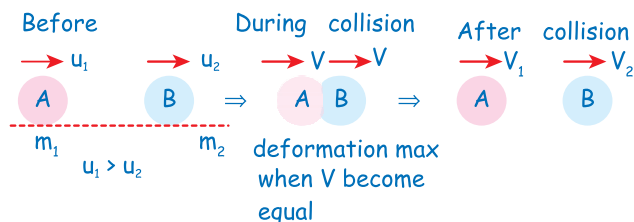
$$\text{Sol. } e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{15 - 7}{10 - 0} = \frac{8}{10} = \frac{4}{5}$$

Q. Find $e = ?$



$$\text{Sol. } e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{2 - (-1)}{2 - (-3)} = \frac{2 + 1}{5} = \frac{3}{5}$$

Elastic Collision



★ Since, $F_{ext} = 0$ so, $\vec{P}_i = \vec{P}_f$

★ Momentum $P \rightarrow$ conserve every time

$$m_1 v_1 + m_2 v_2 = m_1 v + m_2 v = m_1 v_1 + m_2 v_2$$

★ (KE) before collision = (KE) after collision

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

★ Overall no energy loss

$$\star e = 1 = \frac{V_2 - V_1}{U_1 - U_2} = \text{coff of restitution}$$

Inelastic Collision

★ $P \rightarrow$ conserve $0 < e < 1$

★ (K E) $\rightarrow$ loss

★ (KE)_{before collision} > (KE)_{after collision}

Perfectly Inelastic Collision

★ $V_1 = V_2$ (KE)_{loss} $\rightarrow$ max

★ $P \rightarrow$ conserve

★ $e = 0$

देख भाई collision के ques. में data बदल-बदल कर खुब ques. पूछे जाते हैं इसलिए इस ques. की practice अच्छे से कर लेना यहाँ दो eqn. important हैं एक momentum conservation और दूसरी e की definition.

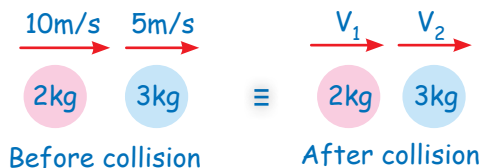


Q. Two ball 2 kg and 3 kg are moving with velocity 10 m/s and 5 m/s as shown in figure. Find velocity of ball after collision if collision is

(a) Elastic ($e = 1$)

(b) Inelastic ($e = 0.5$)

(c) Perfectly inelastic ($e = 0$)



Sol. (a) Elastic ($e = 1$)

$$2 \times 10 + 3 \times 5 = 2V_1 + 3V_2 \quad \dots(i)$$

$$e = 1 = \frac{V_2 - V_1}{5} \quad \dots(ii)$$

Now you can solve both eqn. and get ans.

(b) Inelastic ($e = 0.5$)

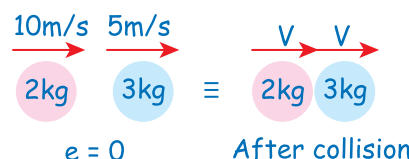
$$2 \times 10 + 3 \times 5 = 2V_1 + 3V_2$$

$$e = \frac{1}{2} = \frac{V_2 - V_1}{5}$$

Now you can solve both eqn. and get ans.

$$\text{loss in (KE)} = (KE)_i - (KE)_f$$

(c) Perfectly inelastic ($e = 0$)



$$P_i = P_f$$

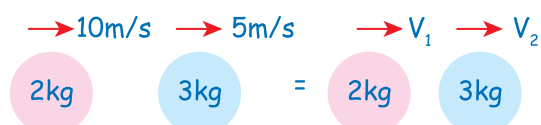
$$2 \times 10 + 3 \times 5 = 2V + 3V$$

$$V = 7$$

$$\text{loss in KE} = (KE)_f - (KE)_i$$

$$= \left[\frac{1}{2} (2+3) 7^2 \right] - \left[\frac{1}{2} \times 2 \times 10^2 + \frac{1}{2} \times 3 \times 5^2 \right]$$

Q. Two balls are moving with is speed 10 m/s and 5 m/s as shown in figure. Find velocity of each ball after collision if collision is elastic ($e = 1$) also find max. potential energy and mini. kinetic energy during collision.



भाई ऐसे ques. में remember दो eqn बहुत use होगी एक momentum conservation की और दूसरी e की definition की

$$\rightarrow \vec{P}_i = \vec{P}_f$$

$$2 \times 10 + 3 \times 5 = 2v_1 + 3v_2 \quad \dots(i)$$

$$e = 1 = \frac{V_2 - V_1}{5} \quad \dots(ii)$$

Solve (1) and (2)

$$V_1 = 4 \text{ and } V_2 = 9$$

During the collision potential energy will be maximum when velocity of both balls become equal.

PE max tab जब दोनों की V same Jitni
(KE) कम उतनी (PE) आ जायगी।
P conserve करे इससे V निकाले और ΔKE
आ जायगी



$$P_i = P_f$$

$$2 \times 10 + 3 \times 5 = 2V + 3V$$

$$V = 7$$

$$(KE)_{\text{before}} = \frac{1}{2} \times 2 \times 10^2 + \frac{1}{2} \times 3 \times 5^2 = \frac{275}{2}$$

$$(KE)_{\text{now}} = (KE)_{\text{min}} = \frac{1}{2} \times 2v^2 + \frac{1}{2} \times 3v^2 = \frac{245}{2}$$

$$\Delta KE = \frac{245}{2} - \frac{275}{2} = -15$$

Hence

$$PE (\text{maximum}) = 15$$

$$(KE)_{\text{min}} = \frac{245}{2}$$

ले भाई एक सवाल और ले and concept बिल्कुल clear हो
जाना चाहिए।



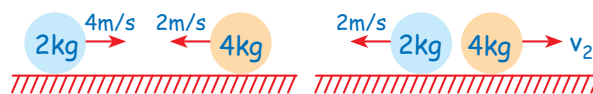
Mogambo khush hua!



Q. Two balls of masses 2 kg and 4 kg are moving towards each other with velocities 4 m/s and 2 m/s respectively on a frictionless surface. After colliding the 2 kg ball returns back with velocity 2m/s. Then find:

- velocity of 4 kg ball after collision
- coefficient of restitution e .
- Impulse of deformation J_D .
- Maximum potential energy of deformation.
- Impulse of reformation J_R .

Just before collision Just after collision



Sol. (a) By momentum conservation,

$$2(4) - 4(2) = 2(-2) + 4(v_2)$$

$$\Rightarrow v_2 = 1 \text{ m/s}$$

$$(b) e = \frac{\text{velocity of separation}}{\text{velocity of approach}} = \frac{1 - (-2)}{4 - (-2)} = \frac{3}{6} = 0.5$$

(c) At maximum deformed state, by conservation of momentum, common velocity is $v = 0$.

$$J_D = m_1(v - u_1) = m_2(v - u_2)$$

$$= 2(0 - 4) = -8 \text{ N-s}$$

$$= 4(0 - 2) = -8 \text{ N-s}$$

$$\text{or } = 4(0 - 2) = -8 \text{ N-s}$$

(d) Potential energy at maximum deformed state U = loss in kinetic energy during deformation.

$$\text{or } U = \left(\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 \right) - \frac{1}{2} (m_1 + m_2) v^2$$

$$= \left(\frac{1}{2} 2(4)^2 + \frac{1}{2} 4(2)^2 \right) - \frac{1}{2} (2 + 4)(0)^2$$

$$\text{or } U = 24 \text{ Joule}$$

$$(e) J_R = m_1(v_1 - v) = m_2(v - v_2)$$

$$= 2(-2 - 0) = -4 \text{ N-s}$$

$$\text{or } = 4(0 - 1) = -4 \text{ N-s}$$

$$\text{or } e = \frac{J_R}{J_D}$$

$$\Rightarrow J_R = e J_D$$

$$= (0.5)(-8) = -4 \text{ N-s}$$

Q. m collides with 2m. Coefficient of restitution is 1. Find V_1 & V_2 if m was moving with velocity V before collision.



$$P_i = P_f$$

$$m \times v + 0 = mv_1 + 2mv_2$$

$$V = V_1 + 2V_2 \quad \dots(1)$$

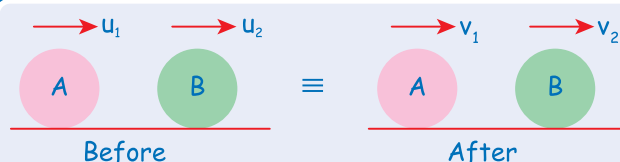
$$e = 1 = \frac{V_2 - V_1}{V} \Rightarrow V_2 - V_1 = V \quad \dots(2)$$

Solve and get

$$v_2 = \frac{2v}{3}, v_1 = \frac{-v}{3}$$

$$\frac{V/3}{\text{green circle}} \quad \frac{2v}{3} \text{ (pink circle)}$$

Fractional decrease in kinetic energy of the colliding particle = 8/9 (do yourself)



$$P_{\text{before}} = P_{\text{after}}$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

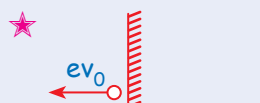
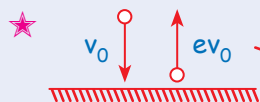
$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

★ If $m_1 = m_2$ and $e = 1$ then after solving we got $v_1 = u_2$ and $v_2 = u_1$

Means if mass is same then in elastic collision ($e = 1$) velocity will exchange



★ If $m_1 \ll m_2$ then $v_1 \approx -eu_1$

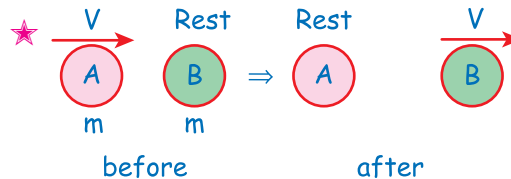
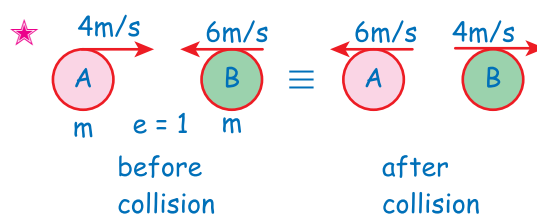


$$|\Delta \vec{P}| = mv_0(1+e)$$

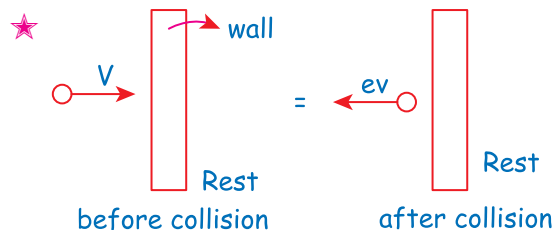
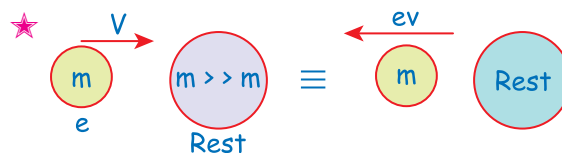
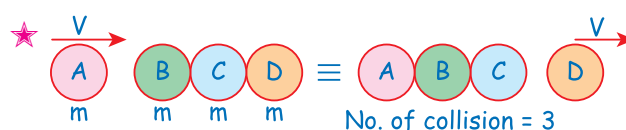
If $e = 1$, $|\Delta \vec{P}| = 2mv_0$



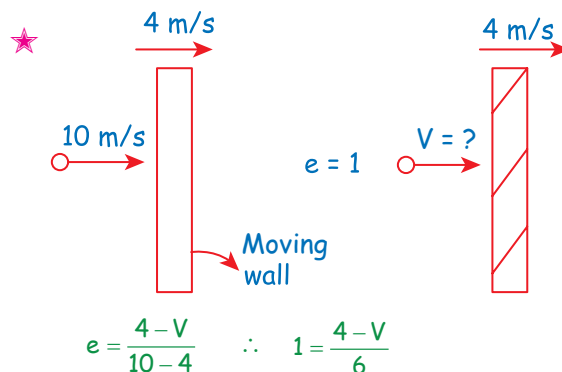
मतलब head on elastic collision में अगर mass same है तो velocity की अदला बदली हो जाएगी और अगर कोई mass v velocity से किसी बहुत भारी mass से टकराएगा तो ev से वापस आएगा



Q. Find velocity of A, B, C, D, after collision. If $e = 1$ for all collision



Q. ball collides with wall moving at $4m/s$ $e = 1$, find V .

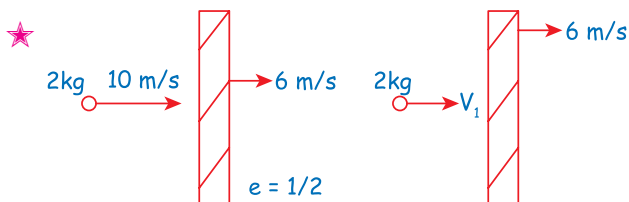


$$e = \frac{4 - V}{10 - 4} \therefore 1 = \frac{4 - V}{6}$$

$$-V = 6 - 4$$

$$V = -2 \rightarrow \text{पीछे}$$

- Q. ball collides with wall moving at 6 m/s $e = 1/2$. Find impulse to ball.



$$e = \frac{6 - V_1}{4} = \frac{1}{2}$$

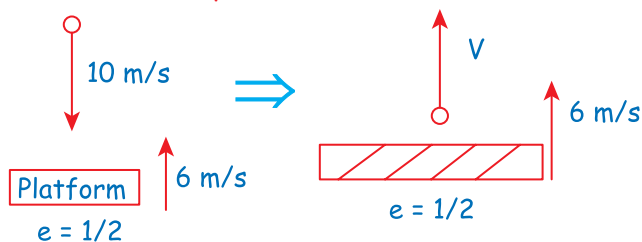
$$V_1 = 4$$

$$(\Delta P)_{\text{ball}} = (2 \times 4) - (2 \times 10) = -12$$

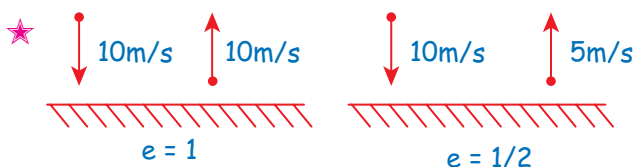
$$\text{Impulse} = -12$$

Wall की velocity पर कुछ खास फर्क नहीं पड़ेगा bcz mass of wall is very large.

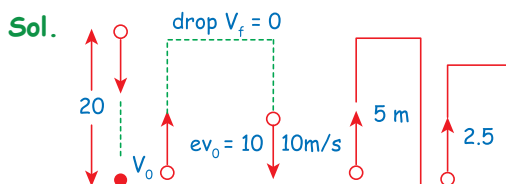
- Q. Ball & platform collide as shown in figure. $e = 1/2$. Find velocity of ball after collision.



$$\text{Sol. } e = \frac{1}{2} = \frac{V - 6}{16} \Rightarrow V = 14$$



- Q. A ball is dropped from 20m height on a floor, $e = 1/2$. Find velocity of ball just after 3rd collision.



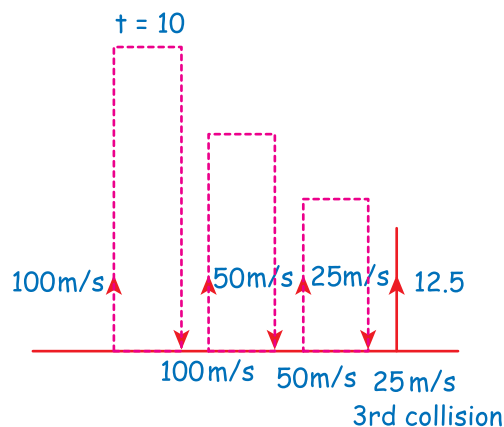
Velocity before hitting floor 1st time

$$V_0^2 = 0^2 + 2 \times 10 \times 20$$

$$V = 20$$

ball जमीन से 20 m/s से टकराएगी और 10 m/s से rebound होगी bcz $e = 1/2$ so, 2nd collision में 5 m/s से rebound होगी और हर collision में उसकी speed half होती जाएगी so, ans = 2.5

- Q. A ball is thrown up with 100m/s as shown. Coefficient to restitution $e = 1/2$

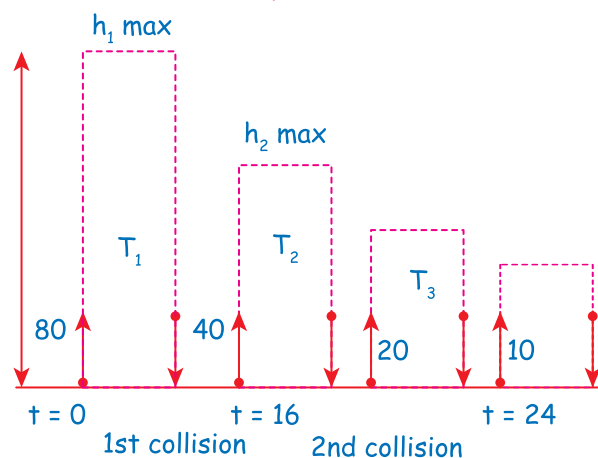


Find when particle will strike floor 3rd time

$$\text{Sol. } (10 + 10) + (5 + 5) + (2.5 + 2.5)$$

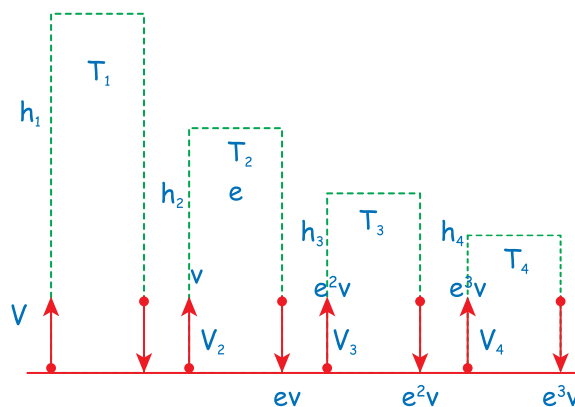
$$t = 26$$

- Q. A ball is thrown up with 80 m/s. $e = 1/2$



- Q. Ball thrown up with velocity V. Coefficient of restitution = e

Sol.



$$V \Rightarrow 1 : e : e^2 = V_1 : V_2 : V_3 \dots$$

$$T \Rightarrow 1 : e : e^2 : e^3 = T_1 : T_2 : T_3 \dots$$

$$h_{\max} \Rightarrow 1 : e^2 : e^4 : \dots =$$

$$T = \frac{2V_y}{g} \quad h_{\max} = \frac{V_y^2}{2g}$$

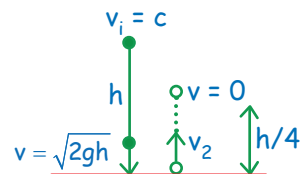
$$(h_1)_{\max} = \frac{v^2}{2g}$$

$$(h_2)_{\max} = \frac{(ev)^2}{2g}$$

$$(h_3)_{\max} = \frac{(e^2v)^2}{2g}$$

- Q.** Two balls of same mass are dropped from the same height h on to the floor. The first ball bounces to a height $h/4$ after the collision & the second ball to a height $h/16$. The impulse applied by the first & second ball on the floor are 1 and Irrespectively. Then

Sol.

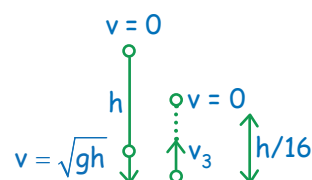


$$0^2 = v_2^2 - 2g \cdot h/4$$

$$\Rightarrow v_2 = \sqrt{gh/2}$$

$$I_1 = \bar{P}_f - \bar{P}_i = m\sqrt{\frac{gh}{2}} + m\sqrt{2gh}$$

$$I_1 = \frac{3m\sqrt{gh}}{\sqrt{2}}$$



$$0^2 = v_3^2 - 2g \frac{h}{16} \Rightarrow v_3 = \frac{\sqrt{gh}}{2\sqrt{2}}$$

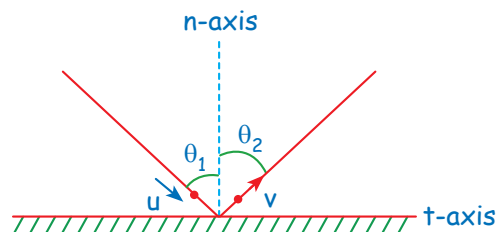
$$\therefore I_2 = m \cdot \frac{\sqrt{2gh}}{4} + m\sqrt{2gh}$$

$$I_2 = \frac{5m\sqrt{2gh}}{4}$$

$$\frac{I_1}{I_2} = \frac{3}{5\sqrt{2}} \Rightarrow 5I_1 = 6I_2$$

Oblique Collision of a Particle with a Fixed Smooth Surface

Consider a ball moving with velocity u at an angle θ_1 with normal to the surface and rebounds with velocity v at angle θ_2 with the normal.



Since the surface is smooth, there is no impulse along the surface. Therefore,

$$u_t = v_t \Rightarrow u \sin \theta_1 = v \sin \theta_2$$

$$v_n = eu_n \Rightarrow eu \cos \theta_2 = v \cos \theta_1$$

$$\Rightarrow \tan \theta_2 = \frac{\tan \theta_1}{e}$$

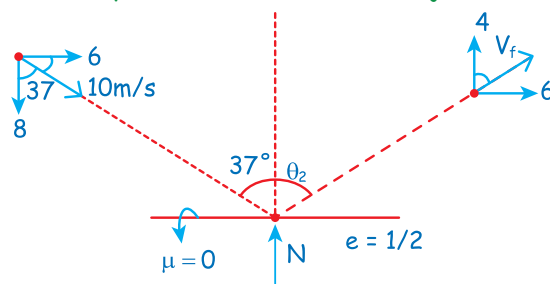
$$v = u \sqrt{\sin^2 \theta_1 + e^2 \cos^2 \theta_1}$$

For elastic collision $e = 1$

$$v = u \text{ and } \theta_2 = \theta_1$$

- Q.** A ball is moving on horizontal plane with velocity 10 m/s strike frictionless surface at an angle 37° with normal to the surface. Find velocity of ball after collision.

Sol. Velocity before collision = $6\hat{i} - 8\hat{j}$



$$\bar{V}_f = 6\hat{i} + 4\hat{j}$$

$$\tan \theta_2 = \frac{6}{4} = \frac{3}{2}$$

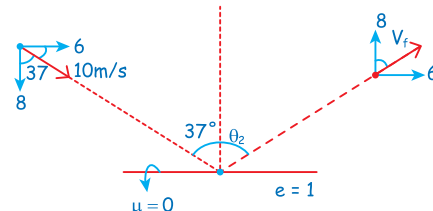
$$(\theta_2 \neq 37^\circ)$$

- Q.** Repeat the above problem if $e = 1$

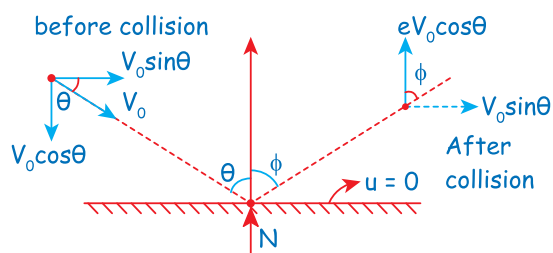
Sol. $v_i = 6\hat{i} - 8\hat{j}$

$$\bar{V}_f = 6\hat{i} + 8\hat{j}$$

$$(\theta_2 = 37^\circ)$$



I hope now you can understand following diagram.



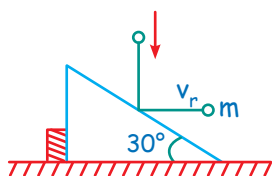
$$\tan \phi = \frac{V_0 \sin \theta}{e V_0 \cos \theta}$$

$$\tan \phi = \frac{\tan \theta}{e}$$

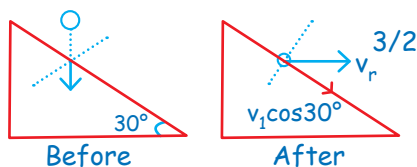
$$\text{If } e = 1 \Rightarrow \phi = \theta$$

$$0 < e < 1 \Rightarrow \phi > \theta$$

- Q. As shown in the figure a body of mass m moving vertically with speed 3 m/s hits a smooth fixed inclined plane and rebounds with a velocity v_f in the horizontal direction. If $\angle$ of inclined is 30° , the velocity v_f will be



Sol. $v_f \cos 30^\circ = \frac{3}{2} \Rightarrow v_f = \sqrt{3} \text{ m/s}$



- Q. A ball strikes a smooth horizontal ground at an angle of 45° with the vertical. What cannot be the possible angle of its velocity with the vertical after the collision. (Assume $e < 1$).

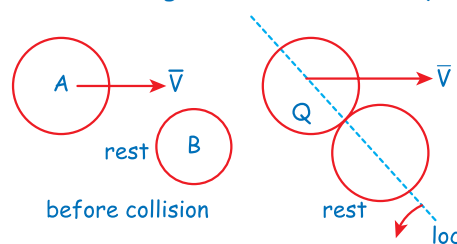


Sol. $\tan \theta = \frac{v_0}{e v_0} = \frac{1}{e} \because e \leq 1$

$$\tan \theta \geq 1 \Rightarrow \theta \geq 45^\circ$$

Oblique Collision

★ When v is not along LOC Collision is oblique collision



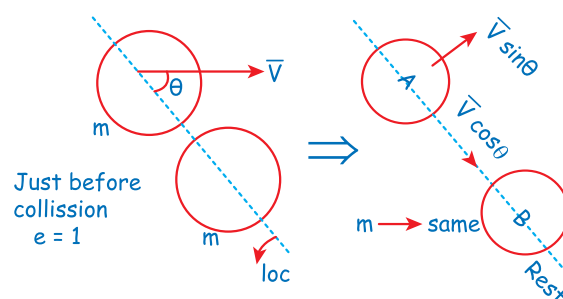
SKC

Oblique collision में सबसे पहले LOC draw करो और collision से पहले दोनों body की velocity के component LOC की तरफ और perpendicular तोड़ लो

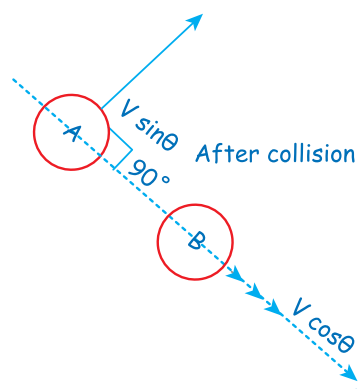
Along the LOC वही पुराने वाले head on collision के concept लगा दो और perpendicular वाली velocity as it is रहने दो बस इतना ही है अब बोलो... ये tough है या आसान



- Q. A ball of mass m moving with velocity V collides with another ball of mass m at rest elastically and obliquely. Find angle between their direction of motion finally.

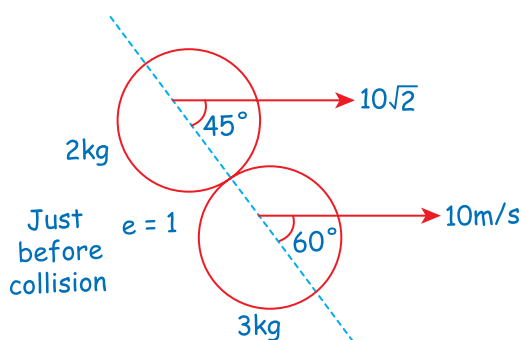


Since, mass is same and collision is elastic so, along the LOC velocity will interchange

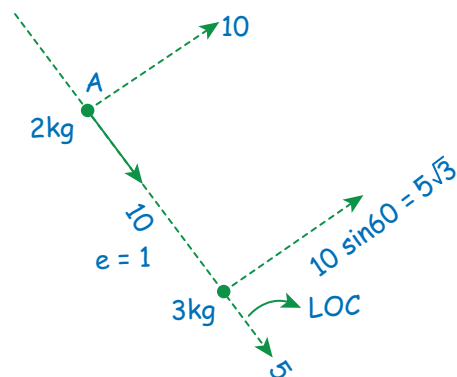


both move in $\perp^r$ direction to each other

Q. Two bodies are colliding as shown in figure. Find final velocities after collision.



Sol.



Momentum conservation along LOC

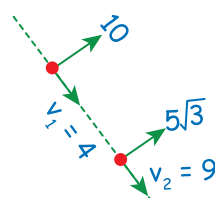
$$P_i = P_f$$

$$20 + 15 = 2v_1 + 3v_2$$

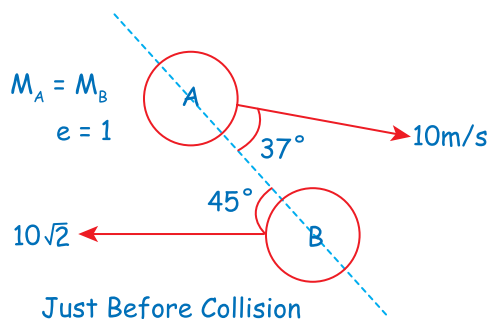
$$e = 1 = \frac{V_2 - V_1}{5}$$

Solve and get

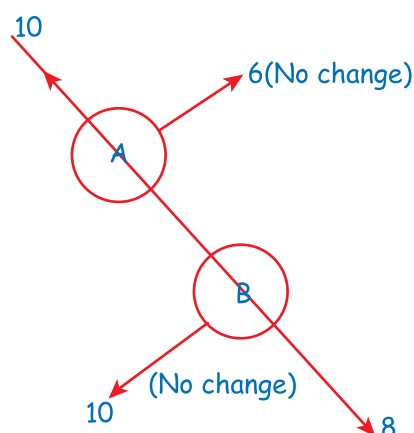
$$V_1 = 4 \text{ \& } V_2 = 9 \text{ so,}$$



Q. Two bodies are colliding as shown in figure. Find velocity of A and B after collision.



Sol.



Impulsive force → Those force which can change the momentum of body by significant amount in very small interval of time

Non-Impulsive force → Those forces which bring very small change in momentum for very small interval of time are called non-Impulsive force.

ex → mg (gravitational force), spring force

Hence we can say

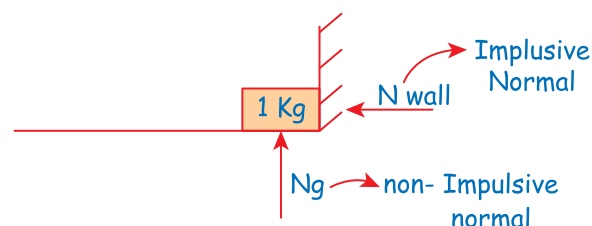
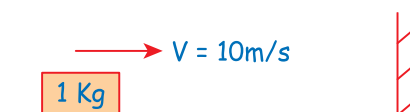
$\vec{P} \rightarrow$ conserve

(1) When net ext force = 0

(2) When non-impulsive force is considered for very small interval of time, momentum will approx. conserve.

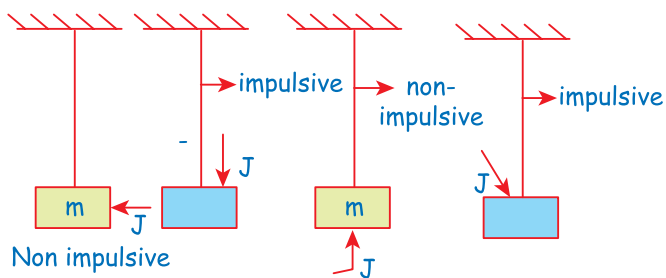
Normal Force

★ Collision की तरफ वाला normal always impulsive

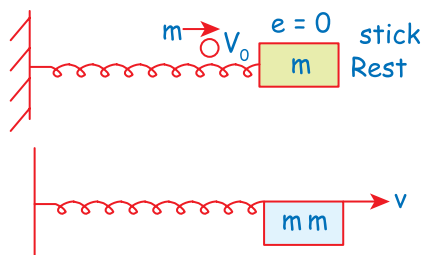


★ If normal is impulsive then corresponding friction will also impulsive $f = \mu N$

Tension



- Q. A particle of mass m moving with velocity V collides with block at rest & stick with it. Find their combined velocity.

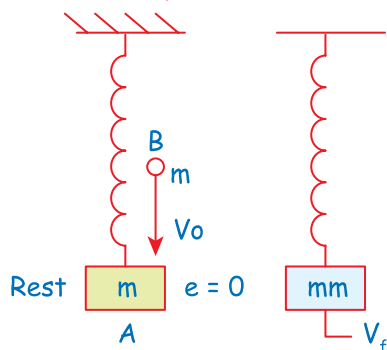


(P_i) just before = (P_f) just after

$$mV_0 + 0 = 2mV_f$$

$$V_f = V_0/2$$

- Q. Particle of mass m collides with block as shown. Find velocity after after collision.



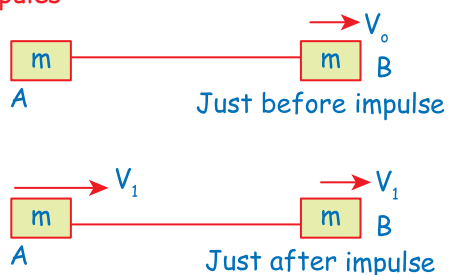
$(A + B) \rightarrow$ system

(P_i) Just before = (P_f) just after collision

$$mV_0 + 0 = 2mV_f$$

$$V_f = V_0/2$$

- Q. Two blocks are connected by a massless string. Block B is given impulse. Find velocity after impulses



$$P_i = P_f$$

$$0 + mV_0 = mV' + mV'$$

$$V' = \frac{V_0}{2}$$

अब हम displacement of COM पर based ques देखेंगे जो कि almost same pattern पर होते हैं।

Concept यह है जिस दिशा में net external force 0 है और COM initially rest पर है तो COM हमेशा rest पर रहेगा and displacement of COM will be 0

If initially com is at rest and $(F_{net})_{ext} = 0$ then

$(\bar{d}_{com})$ displacement of COM = 0

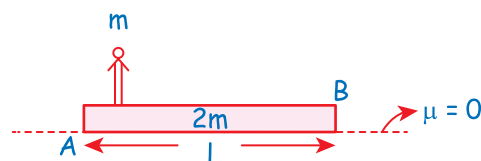
$$\bar{d}_{com} = \frac{m_1 \bar{d}_1 + m_2 \bar{d}_2 + \dots}{m_1 + m_2} = 0$$

$$\Rightarrow m_1 \bar{d}_1 + m_2 \bar{d}_2 = 0$$

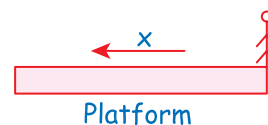


- Q. Initially man and platform both are at rest now man start walking and reaches at B.

Find displacement of platform. Ground is smooth



$$\text{Sol. } m_1 \bar{d}_1 + m \bar{d}_2 = 0$$



Let platform पीछे x गया

$$-2mx + m(l - x) = 0$$

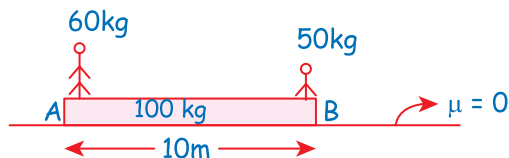
$$L = 3x, \quad x = \frac{L}{3}$$

$$(\bar{d}_{com})_{\text{man wrt ground}} = \frac{2L}{3}$$

अगर मैं platform or man को system मानू तो ext. force is zero. Friction तो है but internal है और initially COM rest पर है So, displacement of COM = 0.



- Q. Initially all are at rest. Both come closer and they reach at center of platform. Find displacement of Platform.



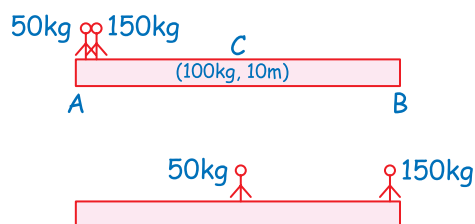
- Sol. मान लो Platform पीछे x गया

$$m_1 \vec{d}_1 + m_2 \vec{d}_2 + m_3 \vec{d}_3 = 0$$

$$-100x + 60(5 - x) - 50(x + 5) = 0$$

$$x = \frac{5}{21}$$

- Q. Initially all are at Rest. Both person are on platform such that 50 kg boy reaches at centre of platform C and 150 kg boy reaches at point B. Find displacement of com of platform



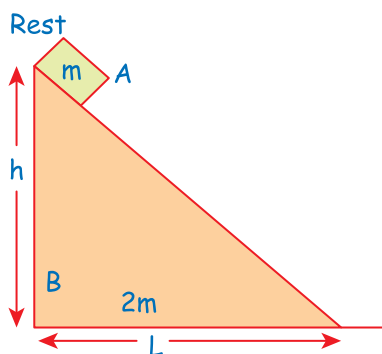
- Sol. मान लो Platform पीछे x गया

$$-100x + 50(5 - x) + 150(10 - x) = 0$$

$$2x + (x + 5) + 3x + 30$$

$$6x = -35 = x = \frac{-35}{6}$$

- Q. A block of mass m is placed on wedge B and released as shown in figure. If all surface are smooth and wedge is free to move find displacement of wedge when block reaches at ground.



यह बहुत important ques है जो बार-बार use होगा

Sol. $(\vec{F}_{net})_x = 0$

Net force along horizontal is zero and initially COM of system is at rest so displacement of COM will be zero along x-axis.

$$m_1 \vec{d}_1 + m_2 \vec{d}_2 = 0$$

मान लो wedge पीछे x गया

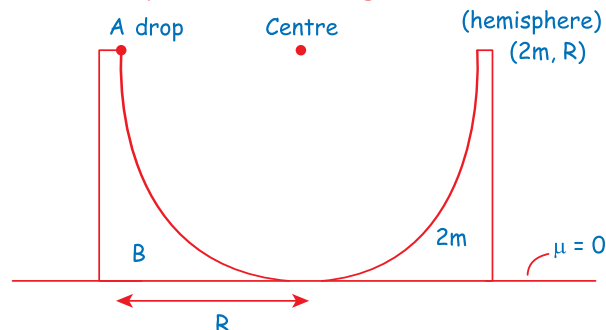
$$(-2mx + m(L - x) = 0) \quad [\text{along horizontal}]$$

$$x = \frac{L}{3}$$

SKC

We carefull overall यहाँ displacement of COM zero नहीं है Bcz जब block नीचे आया तो system का COM भी नीचे आया but x-axis में displacement of mass zero है so, we can say centre of mass system is moving downward in a straight line path till block reaches at ground.

- Q. When ball of mass x reaches at lowest point, find displacement of wedge B.



Sol. $m_1 \vec{d}_1 + m_2 \vec{d}_2 = 0$

मान लो wedge पीछे x गया

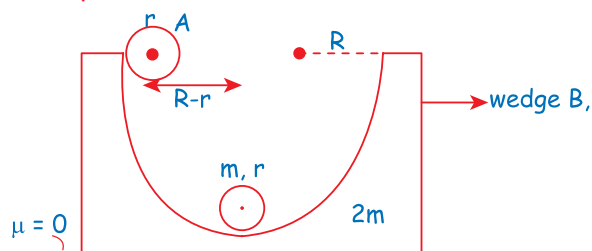
$$(-2mx + m(R - x) = 0) \quad [\text{along horizontal}]$$

$$x = \frac{R}{3}$$

अगर हम ध्यान से देखे तो platform पर man walking, ऊपर के दो ques. और नीचे वाला सवाल भी same pattern पर है बस horizontal distance अलग अलग है बस ये ध्यान रखना हम displacement निकाल रहे है ना की velocity.

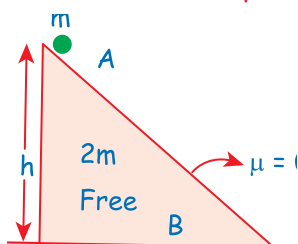


- Q. When sphere A reaches at lowest point find displacement of B.

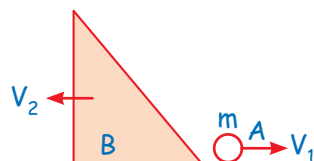


Sol. $-2mx + m(R - r - x) = 0$
 $x = +\frac{(R-r)}{3} \quad (F_{\text{net}})_{\text{ext}} 'x' = 0$

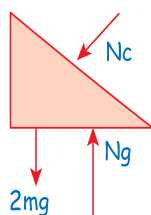
- Q. A ball of mass m is released from point A from a height h on the inclined plane as shown in figure. Find velocity of ball and wedge when ball reaches at lowest point V_1 & V_2 at lowest point



Sol.



$P_i = P_f$ (x मे)
 $0 + 0 = mv_1 - 2mv_2$
 $v_1 = 2v_2$



...(1)



(WET) $W_g + W_N + W_f = \Delta KE$
 $(mgh + 0) + 0 + 0 = \frac{1}{2}mv_1^2 + \frac{1}{2}2mv_2^2$... (2)

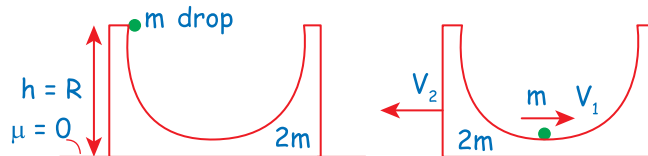
Solve and get V_1 & V_2

Net (WD) by common normal = 0

(WD) by common normal on wedge = $\frac{1}{2} \cdot 2m \cdot v_2^2$

(WD) on ball by normal = $-\frac{1}{2} \times 2mv_2^2$

- Q. A small ball is released as shown in figure. Find velocity of ball and wedge when it reaches lowest point.



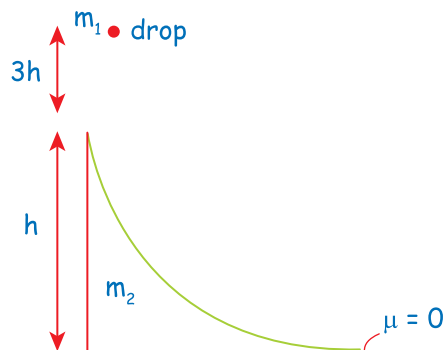
$mv_1 - 2mv_2 = 0$... (1)

$+mgR + 0 + 0 = \frac{1}{2}mv_1^2 + \frac{1}{2}2mv_2^2$... (2)

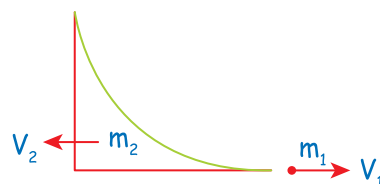
When ball reaches the lowest point N_{ground} will be greater than $3mg$ (सोचो ये कैसे हुआ)



- Q. Find velocity of m_1 when it reaches ground.



Sol.



$P_i = P_f$ (x मे)
 $0 = m_1v_1 - m_2v_2$... (1)

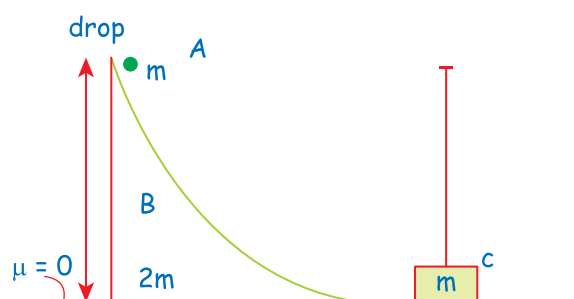
$m_1v_1 = m_2v_2$

$W_g + W_N + W_f = \Delta KE$

$\left(m_1g_1 4h + 0 + 0 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \right)$... (2)

Now you can solve i hope you got concept.

- Q. A ball A is released from an wedge which is free to move. When ball reaches the ground it is strike elastically the block C of same mass as that of ball. Find the minimum value of h so that block C completes vertical circular motion.



देख भाई momentum conservation और WET (kaleen bhaiya) लगा कर A और B की velocity निकाल लेगा ना.....

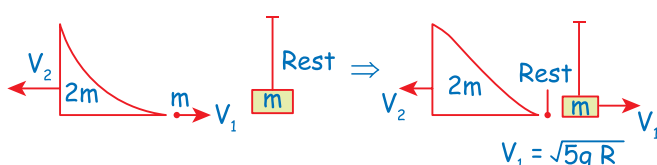
अबे ये तो कितनी बार कर चुके है अब बस A और C का collision कराना है since, $m_A = m_C$ so, velocity exchange तो $v_1 = \sqrt{5gR}$ ही तो करना है..... क्या तुम कर सकते हो।

चलो करके दिखाओ if $m = 2 \text{ kg}$, $R = 2 \text{ m}$

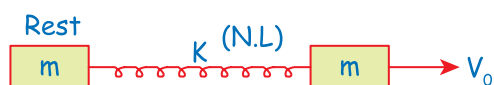
Find h minimum

Ans. $h = 7.5 \text{ m}$

अरे तुम कर तो लोगे ना



Q. For the blocks shown as in figure spring is in natural length



Find max elongation in spring.



At max elongation velocities are equal

initial $\bar{P}_i = P_f$ (final momentum)

$$0 + mv_0 = mv + mv$$

$$v = \frac{v_0}{2}$$

$$W_g + W_N + W_{sp} = \Delta KE$$

$$0 + 0 - \frac{1}{2}k(x_{\max}^2 - 0^2) = \left[\frac{1}{2}m\left(\frac{v_0}{2}\right)^2 \times 2 - \left(\frac{1}{2}mv_0^2\right) \right]$$

$$\frac{-1}{2}kx_{\max}^2 = \frac{-mv_0^2}{4}$$

$$x_{\max} = v_0 \sqrt{\frac{m}{2k}}$$

Q. Find max compression in following figure.

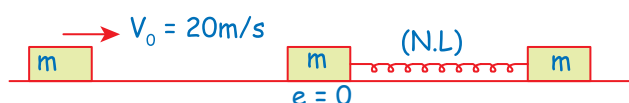


$$P_i = P_f$$

$$0 + mv_0 = mv + mv_0$$

$$x_{\max} = v_0 \sqrt{\frac{m}{2k}}$$

Q. Block with velocity 20m/s collides inelastically with another block find velocity of block just after collision and maximum compression in spring.



$$\text{Sol. } P_i = P_f$$

$$mv_0 + 0 = 2mv \xrightarrow{10 \text{ m/s}}$$

$$v = \frac{v_0}{2} = 10$$



Now you can solve it for to calculate maximum compression or elongation... अरे ये ऊपर जैसा ही तो सवाल हो गया बस दो eqn बनाओ एक momentum conservation और दूसरी WET solve and get.



Q. Find max elongation blocks move as shown in fig.



Sol. Elongation will be maximum when velocity of both the block will be same

$$P_i = P_f$$

$$30 + 120 = 3V + 6V$$

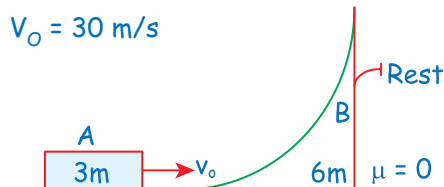
$$V = \frac{150}{9} = \frac{50}{3}$$

$$W_{\text{spring}} = K_f - K_i \text{ (WET)}$$

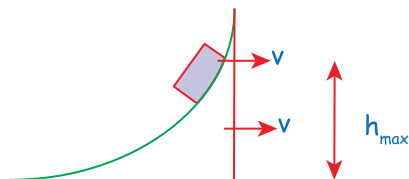
$$-\frac{1}{2}k(x^2_{\text{max}} - 0^2) = \left(\frac{1}{2}(3+6)v^2\right) - \left(\frac{1}{2} \times 3 \times 10^2 + \frac{1}{2} \times 6 \times 50\right)$$
 put value of V , we get

$$x_{\text{max}} = 1$$

Q. Height of B is so large so that block A will not overcome it. Find maximum height reached by block A & maximum speed of B in entire motion. All surface are smooth.



Sol.



$$P_i = P_f \text{ (x मे)}$$

$$3mV_0 = 3mV + 6mV$$

$$V_0 = 30 \text{ m/s} \Rightarrow V = 10$$

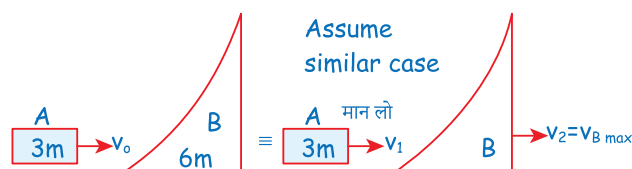
$$Wg + W_{Nc} = \Delta K.E$$

$$-3mgh + 0 = \left(\frac{1}{2}3mv^2 + \frac{1}{2}6mv^2\right)$$

$$-\frac{1}{2}3mv_0^2 - 30 \times h = \frac{3}{2} \times 100 + \frac{6}{2}100 - \frac{3}{2} \times 900$$

$$h = 30 \text{ m} \quad \text{Max height attain by block A}$$

Velocity of B will be maximum when A reaches at ground again



$$P_i = P_f \text{ (x मे)}$$

$$3mV_0 + 0 = 3mV_1 + 6mV_2$$

$$V_0 = V_1 + 2V_2 \quad \dots(1)$$

$$Wg + W_{Nc} + W_{Ng} = \Delta KE$$

$$0 + 0 + 0 = \Delta KE$$

$$K_i = K_f$$

$$\frac{1}{2}3mv_0^2 = \frac{1}{2}3mv_1^2 + \frac{1}{2}6mv_2^2$$

Solve (1) and (2)

$$V_0^2 = V_1^2 + 2V_2^2 \quad \dots(2)$$

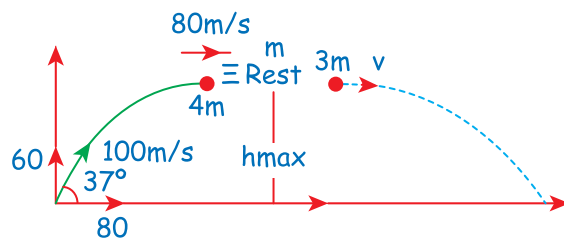
(जब Block Uper जा रहा था तो wedge पर (Nc) लगा जिसका component आगे Ki तरफ आया जिसने Wedge ko speed up किया...



Q. A particle is projected with speed 100 m/s at an angle 37° with the horizontal. At highest point it explode into two parts in the ratio of 1 : 3 such that smaller mass comes to rest. Find

- (1) Velocity of other mass just after explosion.
- (2) Where other mass will strike the ground.

Sol.



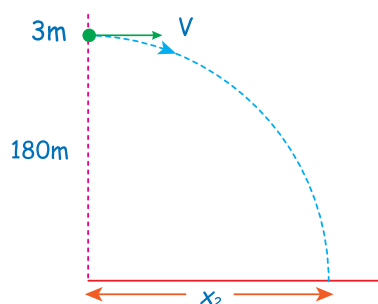
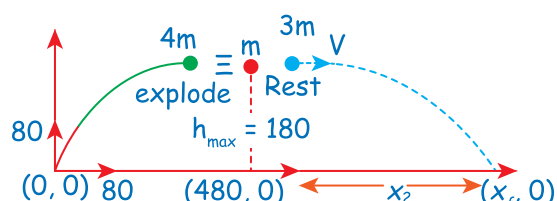
Momentum conservation

(1) $(P_i)_{\text{just before collide}} = (P_f)_{\text{after collide}}$

$$4m \times 80\hat{i} = m \times 0 + 3mV_f$$

$$V_f = \frac{320}{3}$$

$$h_{\max} = \frac{60^2}{2g} = 180$$



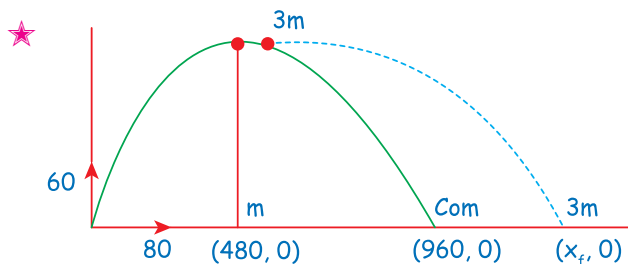
$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 180}{10}} = 6$$

$$x_2 = Vt = \frac{320 \times 6}{3}$$

Co-ordinate of landing point of 3m = $480 + x_2$

$$= 480 + 640 = 1120$$

$$\text{M-II } x_2 = 640$$



$$x_{\text{com}} = \frac{mx_1 + m_2x_2}{m_1 + m_2}$$

$$960 = \frac{m_1 \times 480 + 3m x_f}{4m}$$

$$x_f = \frac{3360}{3} = 1120$$

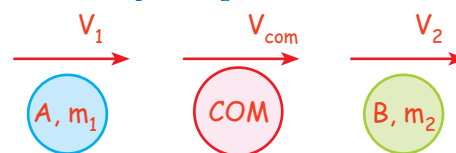
(m, 3m $\Rightarrow$ दोनों एक साथ जमीन पर टकराये)

COM FRAME



COM frame एक फोडू article है जिसके साथ अगर आप खेलना सिख गए तो COM का मजा दुगना हो जाता है कुछ question हम COM frame से बहुत कम step में solve कर लेते हैं लेकिन exam के POV से आप इसे last priority में रखें it's for Adv only (NEET, Mains वाले इसे skip कर सकते हैं)

★ Suppose two ball of mass m_1 and m_2 are moving with velocity V_1 and V_2 as shown in fig.



$$\vec{V}_{\text{com}} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$$

$$\begin{aligned} (\vec{P}_{1/\text{com}}) &\rightarrow m_1\vec{v}_{1/\text{com}} = m_1 \left[\vec{v}_1 - \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} \right] \\ &= m_1 \left[\frac{m_1\cancel{\vec{v}_1} + m_2\vec{v}_2 - m_1\cancel{\vec{v}_1} - m_2\vec{v}_2}{m_1 + m_2} \right] \\ &= \frac{m_1m_2(\vec{v}_1 - \vec{v}_2)}{(m_1 + m_2)} \end{aligned}$$

$$(\vec{P}_{2/\text{com}}) = m_2\vec{v}_{2/\text{com}} = m_2 \left[\vec{v}_2 - \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} \right]$$

$$\frac{m_1m_2(\vec{v}_2 - \vec{v}_1)}{m_1 + m_2}$$

$$\vec{v}_{1/\text{com}} = \vec{v}_1 - \vec{v}_{\text{com}} = \frac{m_2(\vec{v}_1 - \vec{v}_2)}{m_1 + m_2}$$

$$\vec{P}_{1/\text{com}} + \vec{P}_{2/\text{com}} = 0$$

Two Particles System

$$(KE)_{1/\text{com}} = \frac{1}{2}m_1(\vec{v}_{1/\text{com}})^2 = \frac{1}{2}m_1 \left(\frac{m_2(\vec{v}_1 - \vec{v}_2)}{m_1 + m_2} \right)^2$$

$$(KE)_{2/\text{com}} = \frac{1}{2}m_2(\vec{v}_{2/\text{com}})^2 = \frac{1}{2}m_2 \left(\frac{m_1(\vec{v}_2 - \vec{v}_1)}{m_1 + m_2} \right)^2$$

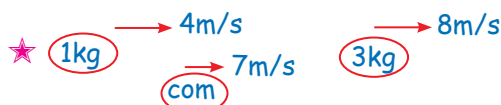
Total KE of system Incom frame

$$= \frac{1}{2} \cdot \frac{m_1 m_2^2 (v_1 - v_2)^2}{(m_1 + m_2)^2} + \frac{1}{2} \cdot \frac{m_2 m_1^2 (v_2 - v_1)^2}{(m_1 + m_2)^2}$$

$$= \frac{1}{2} \cdot \frac{(m_1 m_2)}{(m_1 + m_2)^2} (v_1 - v_2)^2 (m_2 + m_1)$$

$$= \frac{1}{2} \cdot \frac{m_1 m_2}{m_1 + m_2} (v_1 - v_2)^2$$

$\mu = \text{reduced mass}$



$$V_{com} = \frac{1 \times 4 + 3 \times 8}{1 + 3} = \frac{4 + 24}{4} = 7$$

$$\vec{V}_{A/C} = \vec{V}_{A/Com} = (4 - 7)\hat{i} = -3\hat{i}$$

$$(\vec{P}_A)_{incomframe} = 1 \times (-3\hat{i}) = -3\hat{i}$$

$$\vec{V}_{B/C} = (8 - 7)\hat{i} = \hat{i}$$

$$(\vec{P}_B)_{incomframe} = 3 \times 1\hat{i} = +3\hat{i}$$

momentum of system

$$\text{In com frame} = (-3\hat{i}) + (3\hat{i}) = 0$$

In com frame total momentum of system will be always zero

Total K.E of system in ground frame

$$\frac{1}{2} \times 1 \times 4^2 + \frac{1}{2} \times 3 \times 8^2 = 104$$

Total K.E of system in com frame

$$\frac{1}{2} \times 1 \times (3)^2 + \frac{1}{2} \times 3 \times (1)^2 = 6$$

$$\text{Total K.E of COM} = \frac{1}{2} \times 4 \times (7)^2 = 98$$



काम का डब्बा

In com frame Total KE

$$\Rightarrow \frac{1}{2} \mu v_{\text{relative}}^2$$

$$\left(\mu = \frac{m_1 m_2}{m_1 + m_2} \equiv \text{Reduced mass} \right)$$

ये formula
आपको नहीं
भूलना है



For two Particle system

① In com frame total KE of system

$$= \frac{1}{2} \mu v_{\text{relative}}^2 \quad \left(\mu = \frac{m_1 m_2}{m_1 + m_2} \right)$$

② Total KE of system = (KE) of system + (KE)_{com}

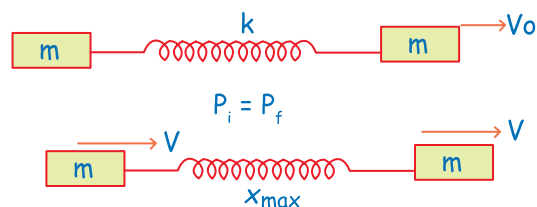
In com frame

$$= \frac{1}{2} \mu v_{\text{rel}}^2 + \frac{1}{2} m_{\text{total}} v_{\text{com}}^2$$

$$(KE)_{\text{system}} = \frac{1}{2} \mu v_{\text{rel}}^2 + \frac{p^2}{2m}$$

$$\text{If } (\vec{F}_{\text{net}})_{\text{ext}} = 0 \Rightarrow \Delta(K.E) = \frac{1}{2} \mu \left[(v_{\text{rel}})_f^2 - (v_{\text{rel}})_i^2 \right]$$

Q. Find max elongation in spring.



Sol.

Method-I

$$0 + mv_0 = mv + mv \Rightarrow v = \frac{v_0}{2}$$

WET

$$-\frac{1}{2} k (x_{\text{max}}^2 - 0) = \frac{1}{2} \cdot m \left(\frac{v_0}{2} \right)^2 \times 2 - \frac{1}{2} m v_0^2$$

$$x_{\text{max}} = v_0 \sqrt{\frac{m}{2k}}$$

Method-II (In COM frame)

$$0 + mv_0 = mv + mv \Rightarrow v = \frac{v_0}{2}$$

WET (work energy theorem in COM Frame)

$$-\frac{1}{2} k (x_{\text{max}}^2 - 0) = \frac{1}{2} \mu \left[(v_{\text{rel}})_f^2 - (v_{\text{rel}})_i^2 \right]$$

$$-\frac{1}{2} k (x_{\text{max}}^2 - 0) = \frac{1}{2} \frac{m}{2} \left[(0)^2 - (v_0)^2 \right]$$

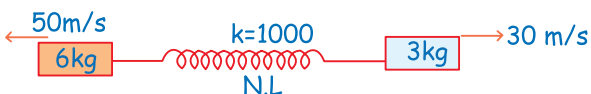
$$x_{\text{max}} = v_0 \sqrt{\frac{m}{2k}}$$

$$\mu = \frac{m \cdot m}{m + m} = \frac{m}{2}$$

हो सकता है कुछ बच्चों को COM frame वाला method tough लगे, believe me अगर अच्छे से practice कि तो this method will save your time जैसे अगले ques में we already know elongation max. तब होगा जब दोनों block की velocity same हो जाएगी means $(V_{rel})_f = 0$. So, direct COM frame में WET में value ही तो put करनी है। जैसे.....



Q. Find max elongation = ?



$$\mu = \frac{6 \times 3}{6 + 3} = 2$$

$$-\frac{1}{2} \times 1000 \times (x_f^2 - 0^2) = \frac{1}{2} \times 2 \times (0^2 - (80)^2)$$

Variable Mass System

Suppose dm is added to m .

$$\begin{matrix} \text{dm} \xrightarrow{V} \\ \text{m} \xrightarrow{u} \end{matrix} = \begin{matrix} \text{m} + \text{dm} \xrightarrow{u + du} \end{matrix}$$

Momentum conservation

$$dm \cdot v + mu = (m + dm)(u + du)$$

$$vdm + \cancel{mu} = \cancel{mu} + mdu + udm + \cancel{dm \cdot du} \quad \text{Neglect}$$

$$Vdm = mdu + udm$$

$$vdm - udm = mdu$$

$$(v - u)dm = mdu$$

$$v_{rel} \frac{dm}{dt} = m \frac{du}{dt}$$

$$\left(\frac{dm}{dt} \right) v_{rel} = ma$$

$$v_{rel} \cdot \frac{dm}{dt} = \text{Force}_{\text{thrust}}$$

Rocket Propulsion

dm is ejected with velocity v_{rel} w.r.t rocket

$$\mu = \frac{m + m_2}{m_1 + m_2} = \frac{m}{2}$$

$$\begin{matrix} \text{m} \xrightarrow{u} \\ \text{m} - \text{dm} \xrightarrow{u + du} \end{matrix} \quad \equiv \quad \frac{dm}{v_{rel}} \geq$$

Momentum conservation

$$P_i = P_f$$

$$mu = dm \cdot V_g + (m - dm)(u + du)$$

[V_g is velocity of dm w.r.t ground]

$$mu = dm \cdot (v_{rel} + u + du) + mu + (mdu - udm - dmdu)$$

$$v_{rel} = v_{\text{gasrocket}}$$

$$v_{rel} = \vec{v}_{\text{gas}} - \vec{v}_{\text{rocket}}$$

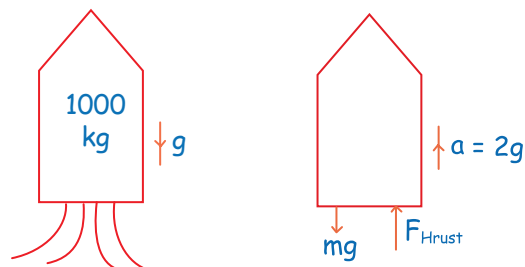
$$mu = v_{rel} \cdot \cancel{dm} + \cancel{u}dm + \cancel{dm}du + \cancel{mu} + mdu - \cancel{u}dm - \cancel{dm}$$

$$-v_{rel} dm = mdu$$

$$-v_{rel} \frac{dm}{dt} = m \frac{dv}{dt} = ma = F_{\text{thrust}}$$

$$F_{\text{thrust}} = -v_{rel} \frac{dm}{dt}$$

★ Gases are ejecting at the rate of 1 kg/sec Find v_{rel} for gas to have an upward acceleration of $2g$ (uplift)



$$F_{\text{thrust}} - mg = ma$$

$$v_{rel} \frac{dm}{dt} - mg = m(2g)$$

$$v_{rel} 1 = 3mg$$

$$v_{rel} = 3 \times 10000 \times 10 = 30000$$

$$\begin{matrix} \text{m} \xrightarrow{u} \\ \text{m} - \text{dm} \xrightarrow{u + du} \end{matrix} \quad \equiv \quad \frac{dm}{v_{rel}} \geq$$

Suppose gas is ejecting at the rate of μ kg/sec

$$\frac{dm}{dt} = \mu = \text{constant}$$

$$dm = \mu dt$$

$$\text{mass of rocket at time 't'} = M = m_0 - \mu t$$

$$m_0 \text{ is the mass of system at } t = 0$$

$$-v_{rel} dm = mdu$$

$$-v_{rel} \mu dt = (m_0 - \mu t)du$$

$$-\int \mu v_{rel} \frac{dt}{m_0 - \mu t} = \int du$$

$$\mu v_{rel} \int_0^t \frac{dt}{m_0 - \mu t} = \int_{u_i}^{v_f} du$$

$$\frac{-\mu v_{rel}}{-\mu} \ln(m_0 - \mu t) \Big|_0^t = v_f - v_i$$

$$V_{rel} [\ln(m_0 - \mu t) - \ln m_0] = v_f - u_i$$

Formula

$$v_f - v_i = v_{rel} \cdot \ln\left(\frac{m_0 - \mu t}{m_0}\right)$$

$$v_f = v_i + v_{rel} \ln\left(\frac{m_0 - \mu t}{m_0}\right)$$

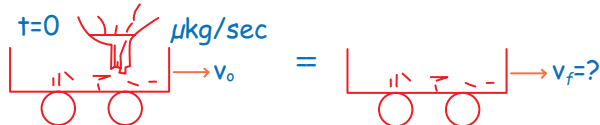
OR

$$= v_i + \ln\left(\frac{m_f}{m_i}\right) \times v_{rel}$$

If $\mu_i = 0$

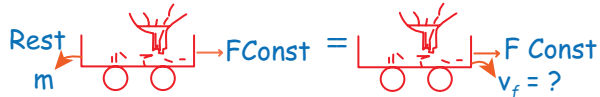
$$v_f = v_{rel} \cdot \ln\left(\frac{m_f}{m_i}\right)$$

Q.



Sol. $mv_o = (m + \mu t) V_f \quad V_f = \frac{mv_o}{m + \mu t} \quad t \uparrow, V \downarrow$

Q.



Sol. Impulse = $\Delta P = P_f - P_i$

$$Ft = (m + \mu t) u_f - 0$$

$$v_f = \frac{Ft}{m + \mu t}$$

अब मैं advance के POV से और आपके concept को और strong करने के लिए कुछ selected question with proper solution and language के साथ attach कर रहा हूँ plz solve them scincerey



Q. A hopper releases grain at a rate dm/dt onto a conveyor belt that moves at a constant speed v . What is the power of the motor driving the belt?

Sol.



Let the system has some arbitrary length of belt whose mass we can call M . This mass of the system increases at the same rate as that the grain falls, so $dM/dt = dm/dt$.

Since the grain falls vertically, $\vec{u} = 0$ and $\vec{v}_{rel} = \vec{u} - \vec{v} = -\vec{v}$, since the speed is constant,

$dv/dt = 0$. Thus from equation for the system of variable mass

$$\vec{F}_{ext} - \vec{v} \frac{dm}{dt} = 0 \Rightarrow \vec{F}_{ext} = \vec{v} \frac{dm}{dt}$$

where $\vec{F}_{ext}$ is the force needed to maintain constant speed because the mass is increasing.

The power required ($P = \vec{F} \cdot \vec{v}$) is

$$P = v^2 \frac{dm}{dt}$$

It is interesting to compare this with the rate at which the kinetic energy of the grain increases.

$$\frac{dK}{dt} = \frac{d}{dt} \left(\frac{1}{2} mv^2 \right) = \frac{1}{2} v^2 \frac{dm}{dt}$$

This is only half the power input. The other half is dissipated as heat when the grain lands on the belt and slips relative to it.

Q. A uniform chain of mass m and length l hangs on a thread and touches the surface of a table by its lower end. Find the force exerted by the table on the chain, when half of its length has fallen on the table. The fallen part does not form heap.

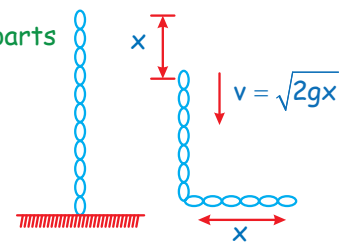
Sol. Let N be the normal reaction acting on the chain.

It consists of two parts

$$N = N_{wt} + N_{thrust}$$

$$N_{wt} = \frac{m}{l} xg$$

$$N_{thrust} = v_{rel} \frac{dm}{dt}$$



If v be the instantaneous velocity of the particle falling on the floor, then

$$v_{rel} = v - 0 = v$$

$$\frac{dm}{dt} = \frac{dm}{dx} \frac{dx}{dt} \text{ or } \frac{dm}{dt} = v \left(\frac{m}{l} \right)$$

$$\text{Thus } N_{th} = v^2 \left(\frac{m}{l} \right), \quad \text{Since } v = \sqrt{2gx}$$

$$\therefore N_{th} = \frac{2mg}{l} x$$

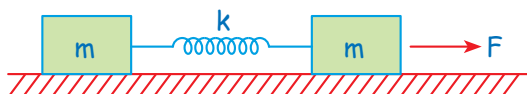
Total reaction is

$$N = \frac{mgx}{l} + \frac{2mgx}{l}, N = 3mg\left(\frac{x}{l}\right)$$

{ $\therefore$ for $x = l/2$ }

$$N = \frac{3}{2}mg$$

- Q.** Two blocks each of mass m , connected by a weightless spring of stiffness k and length L_0 lie on a horizontal surface. A constant force F starts acting on one of the bars (see fig.). Find the maximum distance between the bars during the subsequent motion of the system.



Sol. Consider the system of two blocks and the spring. Force F is external to the system. Therefore neither mechanical energy nor linear momentum will be conserved. Distance between the two blocks will be maximum and minimum when they have the same velocity. Let it be v .

Let d be the distance moved by the right block and x_0 be elongation of the spring when the separation between the blocks is maximum. Consequently, distance moved by the center of mass will be $\left(d - \frac{x_0}{2}\right)$.

The acceleration of the center of mass is given by

$$a_{cm} = \frac{F}{2m}$$

From eqns of constant acceleration, we have

$$v^2 = 2\left(\frac{F}{2m}\right)\left(d - \frac{x_0}{2}\right) \quad \dots (i)$$

and from work-energy theorem, we have

$$\Delta KE + \Delta PE = \text{work done by } F$$

$$\Rightarrow \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2}kx_0^2 = Fd \quad \dots (ii)$$

Solving eqn (i) and eqn (ii), we find $x_0 = F/k$

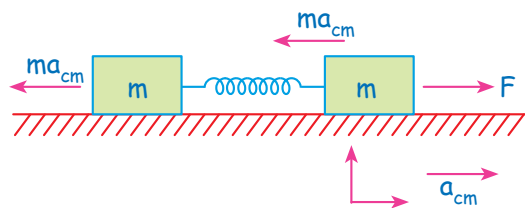
So, the maximum separation is

$$\left(L_0 + \frac{F}{k}\right)$$

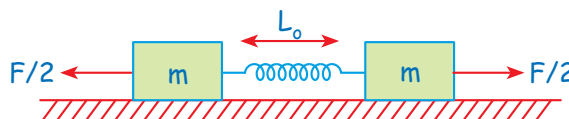
Alternate method(Center of mass frame)

Solving problem in the center of mass frame is always handy as the system is symmetrically placed about the center of mass.

However, a frame fixed to the center of mass in this problem is a non-inertial frame as the acceleration of center of mass is $F/2m$ towards right. Therefore, pseudo forces will act on the blocks towards left.



Setting $a_{cm} = \frac{F}{2m}$, we find that the problem is equivalent to



As the center of mass is at rest in this frame, the blocks move in the opposite direction and come to instantaneous rest at the same time. The extension of the spring is maximum at this instant. Let the two blocks move through distance x_1 and x_2 . The work done on the system by the forces is, therefore

$$W = (F/2)x_1 + (F/2)x_2 = (F/2)(x_1 + x_2)$$

This increases the potential energy of the system. Hence

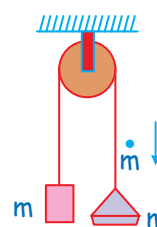
$$(F/2)(x_1 + x_2) = \frac{1}{2}k(x_1 + x_2)^2$$

$$(x_1 + x_2) = F/k$$

The maximum distance between the blocks is, therefore

$$L_0 + x_1 + x_2 = L_0 + F/k$$

- Q.** A block of mass m and a pan of same mass are connected by a string going over a smooth light pulley as shown in figure. Initially the system is at rest until a particle of mass m falls on the pan and sticks to it. If the particle strikes the pan with a speed v find the speed with which the system moves just after the collision.



Sol. Mass m strikes the pan with speed v and stick to it. Let speed of system just after collision is v_1 . For pan and mass m .

Impulse = change in momentum

$$-\int T dt = 2mv_1 - mv$$

For block of mass m

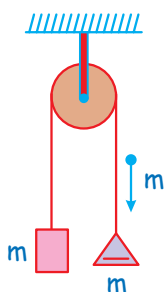
$$\text{impulse} = mv_1$$

$$\int T dt = mv_1$$

$$-mv = 2mv_1 - mv$$

$$mv = 3mv_1$$

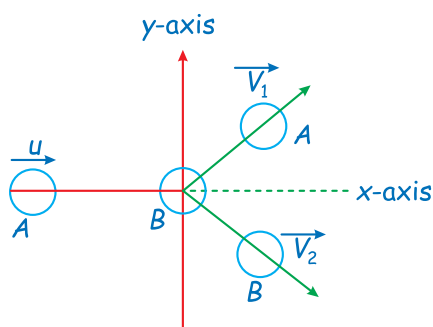
$$v_1 = v / 3$$



Q. A ball A moving with a speed of 9 m/s strikes identical stationary ball B such that after the collision, the direction of each ball makes an angle of 30° with the original line of motion.

(a) Find the speed of the balls after the collision

(b) Is kinetic energy conserved in the collision?



Sol. Choose x-axis along the direction of $\vec{u}$ and y-axis perpendicular to its direction.

Applying conservation of linear momentum along X and Y-axis, we have

$$mu = mv_1 \cos 30^\circ + mv_2 \cos 30^\circ \quad \dots (i)$$

$$\text{and } 0 = mv_1 \sin 30^\circ - mv_2 \sin 30^\circ \quad \dots (ii)$$

After solving eqn (i) and eqn (ii) we find

$$v_1 = v_2 = 3\sqrt{3} \text{ m/s.}$$

(b) Let K_1 and K_2 be the kinetic energy before and after the collision. Then

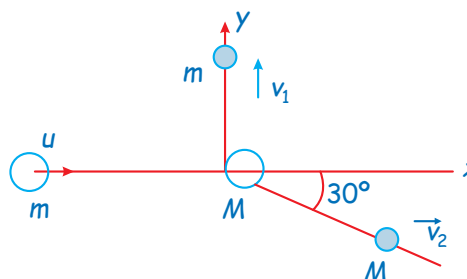
$$K_1 = \frac{1}{2}mu^2 + 0 = \frac{81}{2}m$$

$$\text{and } K_2 = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}m(3\sqrt{3})^2 \times 2 = 27m$$

Since $K_2 < K_1$, kinetic energy of the system is not conserved

Q. A particle of mass m having collided with a stationary particle of mass M deviated by an angle $\pi/2$ whereas particle M recoiled at an angle $\theta = 30^\circ$ to the direction of initial motion of particle m . How much and in what way has the kinetic energy changed after the collision if $\frac{M}{m} = 5$?

Sol.



From conservation of momentum in X and Y direction, we have

$$mu = Mv_2 \cos 30^\circ \quad \dots (i)$$

$$0 = mv_1 - Mv_2 \sin 30^\circ \quad \dots (ii)$$

Solving eqn (i) and eqn (ii) for v_1 and v_2 , we get

$$v_1 = u / \sqrt{3}, v_2 = \frac{m}{M} \frac{2u}{\sqrt{3}}$$

Let K_1 and K_2 be the initial and final kinetic energy of the particles. Then

$$K_1 = \frac{1}{2}mu^2$$

$$\text{and } K_2 = \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2$$

$$= \frac{1}{2}m \left[\frac{1}{3} + \frac{4m}{3M} \right] u^2$$

The percentage loss in K.E. is

$$\frac{K_1 - K_2}{K_1} \times 100 = \frac{2}{5} \times 100 = 40\%$$

Q. A gas molecule having a speed of 300 m/s collides elastically with another molecule of the same mass which is initially at rest. After the collision the first molecule moves at an angle of 30° to its initial direction. Find the speed of each molecule after collision and the angle made with the incident direction by the recoiling target molecule.

Sol. This application corresponds exactly to the situation just discussed, with $m_1 = m_2$, the equations become

$$u_1 = v_1 \cos \theta_1 + v_2 \cos \theta_2 \quad \dots (i)$$

$$v_1 \sin \theta_1 = v_2 \sin \theta_2 \quad \dots (ii)$$

$$\text{and } u_1^2 = v_1^2 + v_2^2 \quad \dots (iii)$$

Squaring and adding the first two equations, we get

$$u_1^2 + v_1^2 - 2u_1 v_1 \cos \theta_1 = v_2^2$$

Combining this with the third equation, we obtain

$$2v_1^2 = 2u_1 v_1 \cos \theta_1$$

$$\text{or (since } u_1 \neq 0) \quad v_1 = u_1 \cos \theta_1 = 260 \text{ m/s}$$

From the third equation, $v_2 = 150 \text{ m/s}$

and finally from the second equation,

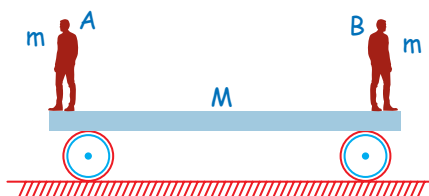
$$\theta_2 = 60^\circ.$$

The molecules move apart at right angles

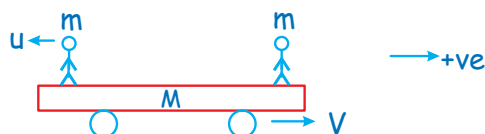
$$(\theta_1 + \theta_2 = 90^\circ).$$

Note: The student should be able to show that in an elastic collision between particles of equal mass, one of which is initially at rest, the recoiling particles always move off at right angles to one another.

- Q.** Two persons A and B, each of mass m are standing at the two ends of rail-road car of mass M . The person A jumps to the left with a horizontal speed u with respect to the car. Thereafter, the person B jumps to the right, again with the same horizontal speed u with respect to the car. Find the velocity of the car after both the persons have jumped off.



- Sol.** When person A jumps, let car achieve velocity V in forward direction. So, velocity of 'A' w.r.t. ground = $(u - V)$ in backward direction.

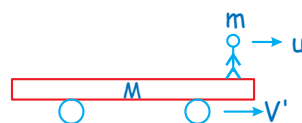


Applying momentum conservation.

$$(M + m)V - m(u - V) = 0, (M + m)V + mV = mu$$

$$V = \frac{m}{M + 2m} u$$

When person B jumps, let velocity of car becomes V' in forward direction.



So velocity of 'B' w.r.t ground = $u + V'$

Applying momentum conservation:

$$(m + M)V = MV' + m(u + V')$$

$$\frac{(m + M)mu}{M + 2m} = (m + M)V' + mu$$

$$V' = \frac{-m^2 u}{(M + m)(M + 2m)}$$

- Q.** A flat car of mass M is at rest on a frictionless floor with a child of mass m standing at its edge. If child jumps off from the car towards right with an initial velocity u , with respect to the car, find the velocity of the car after its jump.

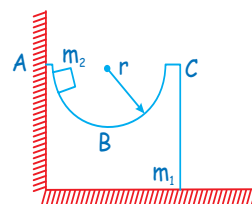
- Sol.** Let car attains a velocity v , and the net velocity of the child with respect to earth will be $u - v$, as u is its velocity with respect to car.



Initially, the system was at rest, thus according to momentum conservation, momentum after jump must be zero, as $m(u - v) = Mv$

$$v = \frac{mu}{m + M}$$

- Q.** A wedge of mass m_1 with its upper surface hemispherical in shape, as shown in Figure, rests on a smooth horizontal surface near the wall. A small block of mass m_2 slides without friction on the hemispherical surface of the wedge. What is the maximum velocity attained by the wedge?



- Sol.** As long as the block moves from A to B, the reaction on the wedge presses it to the wall. When the block reaches the lowermost position, its velocity from energy conservation is

$$v = \sqrt{2gr}$$

When the block moves along the right half of the wedge, during its upward journey as well as downward journey the reaction of the block on the wedge is towards right as shown in figure. Therefore during the entire motion of the block from B to C and C to B, the wedge is accelerated towards right. Thus to find the maximum velocity attained by the wedge at the instant when the block passes separated from the wall,

$$P_i = P_f$$

$$m_2 \sqrt{2gr} = m_1 v_1 + m_2 v_2 \quad \dots (i)$$

From energy conservation, $E_i = E_f$

$$m_2 gr = \frac{m_1 v_1^2}{2} + \frac{m_2 v_2^2}{2} \quad \dots (ii)$$

On solving eqns. (i) and (ii) simultaneously, we obtain two solutions

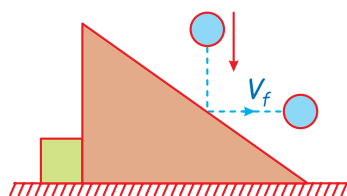
$$v_1 = 0, v_2 = \sqrt{2gr}$$

$$\text{and } v_1 = \frac{2m_2}{m_1 + m_2} (\sqrt{2gr}), v_2 = \frac{m_2 - m_1}{m_1 + m_2} (\sqrt{2gr})$$

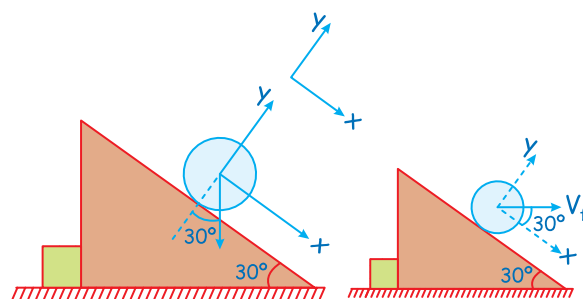
The first solution corresponds to the instant when the block reaches for the first time at point B. At this instant the block moves with velocity v_2 and the wedge is at rest. The second solution corresponds to the instant when the block has the maximum velocity

$$(v_1)_{\max} = \frac{2m_2 \sqrt{2gr}}{m_1 + m_2}$$

- Q.** As shown in the figure a body of mass m moving vertically with speed 3 m/s hits a smooth fixed inclined plane and rebounds with a velocity v_f in the horizontal direction. If angle of inclination is 30° , the velocity v_f will be?



Sol. The momentum along the inclined plane (x-axis) remains conserved. i.e. $\sum p_x = 0$



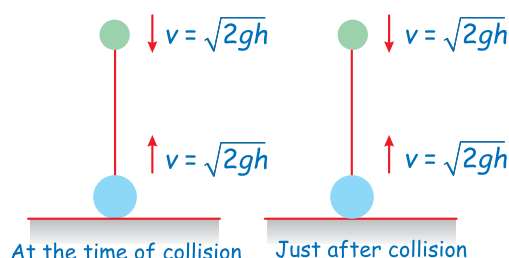
$$\Rightarrow M V_i \sin 30^\circ = M V_f \cos 30^\circ$$

$$\Rightarrow V_f = V_i \frac{\sin 30^\circ}{\cos 30^\circ} = V_i \tan 30^\circ$$

$$\Rightarrow V_f = V_i \frac{1}{\sqrt{3}} = 3 \cdot \frac{1}{\sqrt{3}} = \sqrt{3} \text{ m/s}$$

- Q.** A small ball of mass m is connected by an inextensible massless string of length l with another ball of mass $M = 4m$. They are released with zero tension in the string from a height h as shown in figure. Find the time when the string becomes taut for the first time after the mass M collides with the ground. Take all collisions to be elastic.

Sol. Just after collision velocities of both the balls will be $\sqrt{2gh}$ in opposite directions. Relative acceleration between the two balls is zero and relative velocity of approach is $2\sqrt{2gh}$. Hence they will collide after a time



$$t = \frac{l}{2\sqrt{2gh}} \quad \dots (i)$$

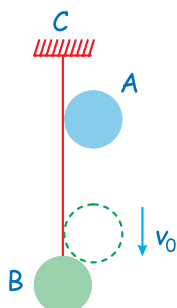
At the time of collision of two balls relative velocity of approach = $2\sqrt{2gh}$

Hence relative velocity of separation will also be $2\sqrt{2gh}$ (collision is elastic). Hence the string becomes tight after the same time

$$t = \frac{l}{2\sqrt{2gh}}$$

Hence the total time will be $2t$ or $\frac{l}{\sqrt{2gh}}$

- Q. Ball B is hanging from an inextensible cord BC. An identical ball A is released from rest when it is just touching the cord and acquires a velocity v_0 before striking ball B. Assuming perfectly elastic impact ($e = 1$) and no friction, determine the velocity of each ball immediately after impact.

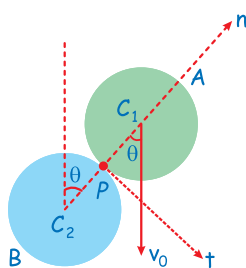


Sol. Ball A is free to move in a plane (after collision). So its velocity can be resolved in two mutually perpendicular directions. Let us resolve it along common tangent and common normal directions. Let v_t and v_n be the corresponding components of ball 'A' just after collision in these two directions. Ball B is attached to a vertical string. So, just after collision its velocity will be horizontal. Let it be v .

We have $\theta = 30^\circ$

- (i) Velocity component along common tangent direction remains unchanged. Hence,

$$u_t = v_t = v_0 \sin \theta = \frac{v_0}{2} \quad \dots (i)$$



Just before collision

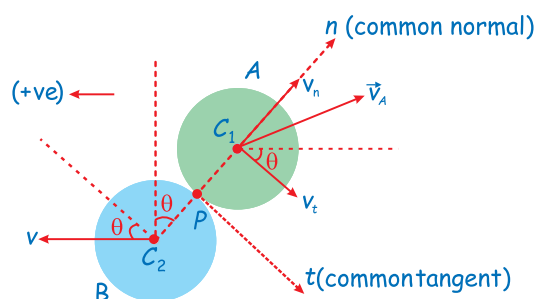
- (ii) Linear momentum along horizontal will remain unchanged. Hence,

$$mv - mv_t \cos 30^\circ - mv_n \sin 30^\circ = 0$$

$$\text{or } v - \frac{\sqrt{3}v_t}{2} - \frac{v_n}{2} = 0 \quad \dots (ii)$$

- (iii) $e = \frac{v_{\text{sep.}}}{v_{\text{app.}}} = 1$ ($\because$ elastic collision)

$$\text{i.e. } \frac{v_n + v \sin \theta}{v_0 \cos \theta} = 1$$



Just after collision

$$\therefore v_n + v \sin \theta = v_0 \cos \theta$$

$$\text{or } v_n + \frac{v}{2} = \frac{\sqrt{3}v_0}{2} \quad (\text{as } \theta = 30^\circ) \quad \dots (iii)$$

Solving Eqs. (i), (ii) and (iii), we get

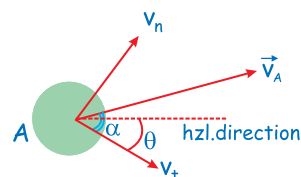
$$v = 0.693v_0, \quad v_n = 0.52v_0 \quad \text{and} \quad v_t = 0.5v_0$$

$\therefore$ Velocity of 'B' after collision $v = 0.693v_0$ (horizontally)

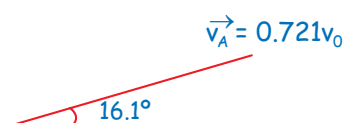
and Velocity of 'A' after collision

$$v_A = \sqrt{v_n^2 + v_t^2} = 0.721v_0$$

$$\alpha = \tan^{-1} \left(\frac{v_n}{v_t} \right) = 46.1^\circ$$



Hence, v_A makes an angle of $(46.1^\circ - 30^\circ)$ or 16.1° with horizontal as shown



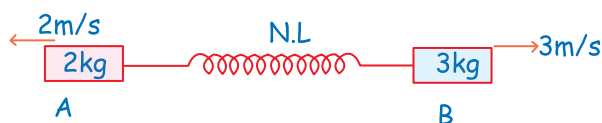
(Vcom)

wrt com $\Rightarrow V_A V_B$

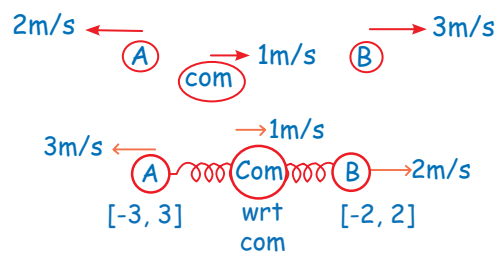
wrt ground $\Rightarrow V_A V_B$ (Vcom add $\checkmark$)

जमीन पर उतरे

- Q. Find max and min velocity of A and B in entire motion



$$\text{Sol. } V_{\text{com}} = \frac{3 \times 3 - 2 \times 2}{2 + 3} = 1$$



$$V_B \text{ Range} = [-2, 2]$$

$$V_A \text{ Range} = [-3, 3] \text{ wrt com}$$

$$\vec{V}_{A/C} = \vec{V}_A - \vec{V}_C$$

$$\vec{V}_{A/\text{ground}} = \vec{V}_{A/C} + \vec{V}_C / g$$

$$[-3 + 1, 3 + 1] \text{ wrt ground } [-2 + 1, 2 + 1]$$

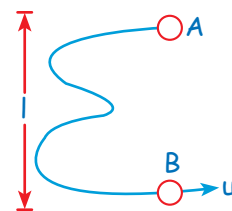
$$[-2, 4] \quad \quad \quad [-1, 3]$$

$$(\vec{V}_A)_{\max} = 4\hat{i}, (\vec{V}_B)_{\max} = 3\hat{i}$$

$$(\vec{V}_A)_{\min} = 0, (\vec{V}_B)_{\min} = 0$$

- Q. Two particles A and B of equal mass m each are attached by a string of length $2l$ and initially placed over a smooth horizontal table in the position shown in figure. Particle B is projected across the table with speed u perpendicular to AB as shown in figure. The ratio $\frac{v_B}{v_A}$ after the string becomes taut is $\sqrt{\frac{\alpha}{\beta}}$. Find the least integral value of $(\alpha + \beta)/2$.

(where v_B and v_A are speed of B and A respectively after the string become taut.)



$$\text{Sol. } m u \cos 30^\circ = 2 \times m v_x$$

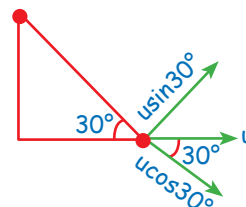
$$v_x = \frac{\sqrt{3}u}{4}$$

$$\Rightarrow v_A = \frac{\sqrt{3}u}{4}$$

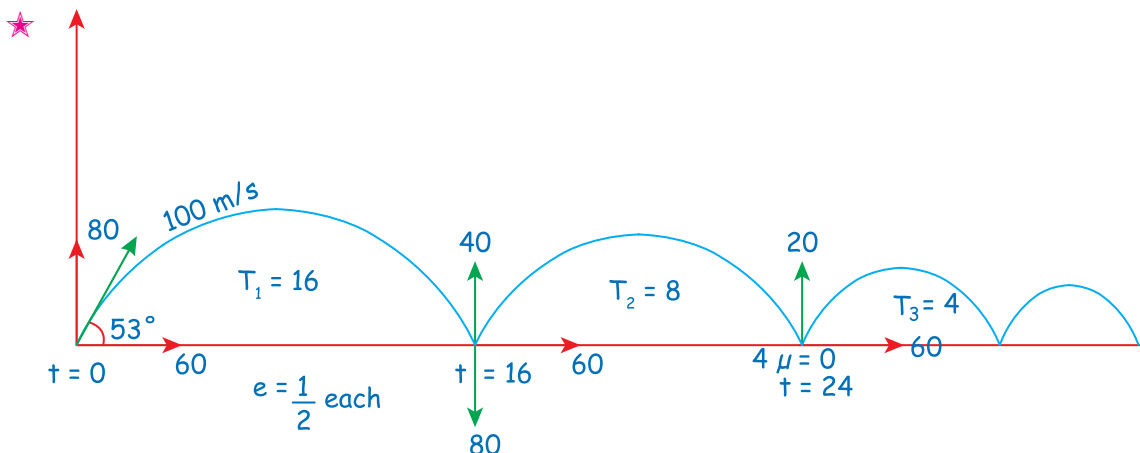
$$\Rightarrow v_B = \sqrt{\frac{3u^2}{16} + \frac{u^2}{16}} = \frac{u}{4} \times \sqrt{7}$$

$$\frac{v_B}{v_A} = \sqrt{\frac{7}{3}} = \sqrt{\frac{\alpha}{\beta}}$$

$$\frac{\alpha + \beta}{2} = 5$$



- Q. A ball is thrown at 100 m/s at 53° from horizontal. Coefficient of restitution $e = 1/2$ between ball and ground.



$$(h_2)_{\max} = \frac{(40)^2}{2g}$$

$$T_1 = 16, T_2 = 8, T_3 = 4$$